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Design of a Low-Power 12-bit Binary SAR-ADC Using the IHP Open PDK

Masterarbeit im Fach Informations- und Kommunikationstechnik

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Design of a Low-Power 12-bit Binary SAR-ADC Using the IHP Open PDK

Masterarbeit im Fach Informations- und Kommunikationstechnik

vorgelegt von

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(Muhammad Fawad Pervez Siddiqui)

Erlangen, 22. Juli 2026

Abstract

Successive-approximation-register (SAR) analog-to-digital converters (ADCs) are widely used where moderate resolution must be achieved at low power, and their largely capacitive-and-comparator structure makes them well suited to implementation on open process design kits and open-source design tools. This thesis presents the design of a 12-bit SAR ADC in the IHP SG13G2 130 nm open PDK, carried out entirely with a free and open-source toolchain.

The converter is fully differential and uses monotonic switching. It comprises a split 6+6 binary-weighted metal–insulator–metal capacitor array, a bootstrapped sampling switch for linear input sampling, a comparator consisting of a resistively-loaded preamplifier followed by a StrongARM latch, and a synchronous SAR controller described in a hardware description language and implemented through an automated digital flow. The design choices at each block are motivated against the requirements of a 12-bit converter, and the analog blocks are laid out with attention to device matching. The complete converter passes design-rule and layout-versus-schematic verification.

Pre-layout simulation, full parasitic extraction being outside what the open-source flow could produce for a design of this size, gives an effective resolution of 11.3 bits, a signal-to-noise-and-distortion ratio of 69.5 dB, peak differential and integral nonlinearity of 0.41 LSB and 1.16 LSB, and a total power of 272 μ W at 2 MS/s, corresponding to a Walden figure of merit of 54 fJ per conversion step. The work demonstrates that a converter of this resolution can be designed, laid out, and verified end to end on an open PDK using only freely available tools.

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Chapter 1

Introduction

1.1 Motivation

Analog-to-digital converters (ADCs) are among the most fundamental building blocks in modern integrated circuits. Nearly every mixed-signal system requires an interface between the continuous analog world and the digital domain, making the data converter an essential component across applications ranging from sensor front-ends and biomedical devices to communication systems and instrumentation [1]. Among available architectures, the successive-approximation-register (SAR) ADC is particularly attractive for medium-resolution, medium-speed applications, offering an excellent balance of resolution, power efficiency, and silicon area [1].

In recent years, open-source process design kits (PDKs) have begun to transform the landscape of integrated circuit design. By removing the licensing restrictions and non-disclosure agreements traditionally associated with commercial technologies, open PDKs make silicon design accessible to a far broader community, including universities, independent researchers, and students. The IHP SG13G2 process is a recent addition to this ecosystem: a 130 nm SiGe BiCMOS technology released as a fully open PDK [2]. As a relatively new platform, it still lacks the breadth of openly available reference designs that more established open PDKs have accumulated, particularly in the analog and mixed-signal domain where designs are inherently harder to port and reuse than digital logic.

This gap provides the primary motivation for this work. Resources such as the IHP Analog Academy [3] offer a useful entry point for designers new to this technology, but openly available reference designs remain scarce. Given the central role of data converters in mixed-signal systems, a thoroughly documented ADC design would represent a valuable contribution to the SG13G2 ecosystem. By developing a fully differential 12-bit SAR ADC in SG13G2 and making the complete design openly available, this thesis aims to lower the barrier of entry for future designers and

to contribute a reusable, well-documented building block to its growing design community.

A second motivation concerns the design tools themselves. Commercial IC design tools are expensive and largely inaccessible to students and independent designers, raising the barrier to entering the field of silicon design. This work is therefore carried out entirely using free, open-source tools spanning schematic capture, simulation, layout, physical verification, and digital synthesis. Completing a full converter design in this way also serves as a practical demonstration that open-source tools have matured sufficiently for realistic analog and mixed-signal design, and not only for digital or purely academic examples.

1.2 The IHP SG13G2 Open PDK

All design work in this thesis targets the IHP SG13G2 process, a 130 nm SiGe BiCMOS technology from IHP (The Leibniz Institute for High Performance Microelectronics). Its Process Design Kit (PDK) was released as open source, making it the first European open-source PDK and one of only a small number of openly available foundry technologies worldwide [2]. While the technology also provides high-speed SiGe:C bipolar transistors ($f_T = 300$ GHz) aimed at RF applications, these are not used in this work; the converter is implemented entirely with the CMOS devices [4].

For the design presented here, the relevant features of the process are its 1.2 V thin-oxide core transistors, from which the entire ADC is built, and its metal–insulator–metal (MIM) capacitors, which form the capacitive DAC described in later chapters. The backend provides five thin and two thick aluminium metal layers for routing, together with a dedicated MIM layer [4]. The PDK ships with the device models, symbols, and design-rule and layout-versus-schematic decks required by the open-source tool flow, so a design can be simulated, laid out, and verified without any proprietary software. It should be noted that the open-source PDK is, at the time of writing, still described as a preliminary release rather than a production-qualified kit [5]; since this thesis is concerned with demonstrating a complete design and verification flow rather than fabricating a production part, this does not affect the work.

1.3 Successive-Approximation Register ADCs

A successive-approximation-register (SAR) ADC converts an analog input to a digital code using a binary-search algorithm. The input is first sampled onto a capacitor, after which the converter resolves the result one bit at a time, from the most significant bit (MSB) to the least significant bit (LSB). At each step a comparator decides whether

the sampled input is above or below a reference level generated by a digital-to-analog converter (DAC); the SAR logic uses this decision to set or clear the corresponding bit and updates the DAC accordingly for the next step [1]. For an N -bit converter, the conversion therefore requires N comparison cycles.

This architecture is well suited to medium-resolution, medium-speed applications. Because it contains only a single comparator, a DAC, and digital logic, and no power-hungry active amplifiers in the conversion path, the SAR ADC is inherently power- and area-efficient, which is why it is widely used in sensor interfaces and other low-power systems [1]. In this work the DAC is implemented as a charge-redistribution capacitive DAC, and the whole converter is built in a fully differential configuration; these choices, along with the switching scheme and comparator design are described in detail in Chapter 2 and Chapter 4.

1.4 Design Goals

The objective of this thesis is the complete design, layout, and verification of a SAR ADC in the IHP SG13G2 open-source technology, carried out entirely with open-source tools and released as an openly available design. The concrete target specifications are summarised in Table 1.1.

Table 1.1 – Target specifications of the SAR ADC.

Parameter	Target
Resolution	12 bit
Architecture	Fully differential, charge-redistribution SAR
Sampling rate	2 MS/s
Supply voltage	1.2 V
Technology	IHP SG13G2 (130 nm BiCMOS)
Tool flow	Fully open-source
Deliverable	Verified, layout-level hardened macro

The main contributions of this work are: (i) a complete, fully differential 12-bit SAR ADC designed and laid out in the SG13G2 open-source PDK; (ii) an end-to-end open-source design and verification flow spanning schematic capture, simulation, layout, physical verification, and digital synthesis; and (iii) an openly available, documented reference design.

1.5 Thesis Outline

The remainder of this thesis is organised as follows: Chapter 2 introduces the theoretical background required for the design, including the operating principle of

SAR converters, charge-redistribution DACs, and the relevant performance metrics. Chapter 3 reviews the open-source PDKs, tool flows, and SAR ADC techniques on which this work builds, and positions it against recent implementations. Chapter 4 presents the overall architecture of the converter and the reasoning behind the main design choices. Chapter 5 describes the circuit- and layout-level implementation of each block, together with the open-source design and verification flow. Chapter 6 presents the simulation and verification results, both at schematic level and after layout extraction. Finally, Chapter 7 summarises the work and outlines directions for future development.

Chapter 2

Background

This chapter introduces the concepts required to follow the design presented in the later chapters. It begins with the operating principle of the successive-approximation-register (SAR) ADC, then describes the charge-redistribution capacitive DAC and the switching schemes relevant to this work, and finally defines the performance metrics used throughout the thesis to evaluate the converter.

2.1 Operating Principle of the SAR ADC

The SAR ADC determines the digital equivalent of an analog input by performing a binary search over the available output codes. A conversion begins by sampling the input onto a sampling capacitor. The converter then resolves the result one bit at a time, starting from the most significant bit (MSB). At each step, a comparator compares the held input against a voltage produced by an internal digital-to-analog converter (DAC). Based on the comparator's decision, the SAR logic either keeps or clears the bit currently under test and updates the DAC output for the next comparison [1]. After N such steps, the N -bit output code is available.

Figure 2.1 shows the basic building blocks common to this architecture: a sample-and-hold circuit, a comparator, a DAC, and the SAR control logic. The absence of any high-gain analog amplifier in the conversion loop is the key reason for the SAR ADC's low power consumption, since the only active analog element is the comparator, which is typically implemented as a dynamic circuit that draws no static current. This makes the architecture particularly well suited to low-power, medium-resolution applications such as sensor front-ends and biomedical devices [1].

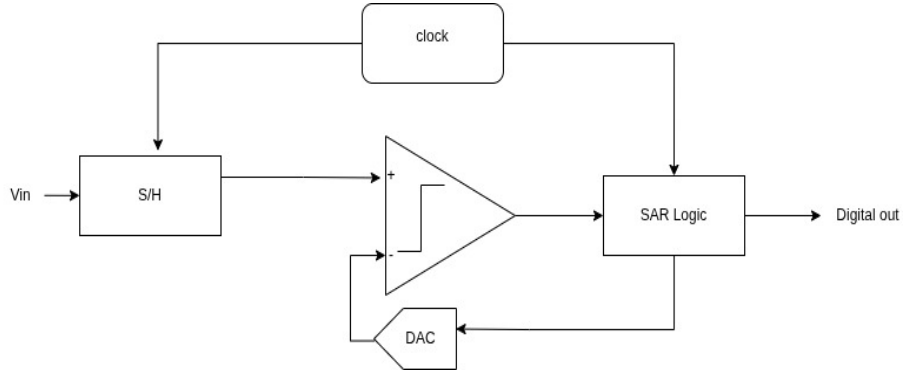


Figure 2.1 – Block diagram of the successive-approximation-register ADC. The sample-and-hold captures the differential input, and the comparator compares it against the voltage produced by the capacitive DAC. The SAR logic uses each comparator decision to update the DAC for the next step and to assemble the N -bit output code.

2.2 Quantization and the Ideal Converter

Before turning to the circuit, it is worth stating the limit that no converter can escape. An ADC maps a continuous input range onto a finite set of 2^N output codes, so every conversion rounds the input to the nearest code and discards the remainder. This rounding is called *quantization*, and the smallest step it can resolve is one least-significant bit,

$$V_{\text{LSB}} = \frac{V_{\text{FS}}}{2^N}, \quad (2.1)$$

where V_{FS} is the full-scale input range and N the resolution. Even a perfect converter, free of any circuit non-ideality, still commits a quantization error e equal to the difference between the input and the code it is assigned to. For a busy input this error is well modelled as a random variable spread uniformly over the interval $[-V_{\text{LSB}}/2, +V_{\text{LSB}}/2]$, which gives it a mean-square value

$$\overline{e^2} = \frac{1}{V_{\text{LSB}}} \int_{-V_{\text{LSB}}/2}^{+V_{\text{LSB}}/2} e^2 \, de = \frac{V_{\text{LSB}}^2}{12}. \quad (2.2)$$

Treating this quantization error as a noise source sets the best possible signal-to-noise ratio of the converter. For a full-scale sinusoidal input, whose power is $V_{\text{FS}}^2/8$, dividing the signal power by the quantization-noise power of Equation (2.2) and expressing the result in decibels yields the familiar limit

$$\text{SNR}_{\text{ideal}} = 6.02N + 1.76 \text{ dB}. \quad (2.3)$$

Equation (2.3) is the reference against which a real converter is judged: it says that each additional bit of resolution buys roughly 6 dB of dynamic range, and it underlies the effective-number-of-bits metric defined in Section 2.9. Every mechanism

described in the remainder of this chapter, capacitor mismatch, switching behaviour, and sampling noise acts to keep the real converter as close to this ideal as possible.

2.3 Charge-Redistribution Capacitive DAC

In this work, the DAC is implemented as a charge-redistribution capacitive array, which is the dominant DAC realisation in modern SAR converters. A binary-weighted array of capacitors performs three functions simultaneously: it samples the input, holds the sampled value, and generates the reference levels used during the binary search. Because the same capacitors are reused for both sampling and digital-to-analog conversion, no separate sample-and-hold amplifier is required, which reduces area, power, and complexity [1].

The operating principle can be understood from charge conservation. Consider an N -bit array of binary-weighted capacitors $2^{N-1}C, \dots, 4C, 2C, C$ together with a terminating unit capacitor, giving a total capacitance

$$C_{\text{tot}} = 2^N C, \quad (2.4)$$

where C is the unit capacitance. During the sampling phase, the input voltage V_{in} is applied across the array and the stored charge is proportional to V_{in} . During conversion, the bottom plate of each capacitor is driven from a reference voltage under control of the SAR logic. Switching a capacitor C_i redistributes charge within the array and shifts the voltage at the comparator input node by an amount proportional to the ratio C_i/C_{tot} . Because the capacitors are binary-weighted, each successive bit decision moves the DAC voltage by half the previous step, which is exactly the behaviour required to perform a binary search over the input range [1]. The particular sequence in which the capacitors are driven—the switching scheme—determines how much energy this process costs and is treated separately in Section 2.5.

The accuracy of this process depends directly on how precisely the capacitor ratios are realised. Any mismatch between capacitors that should be in a 2 : 1 ratio translates into a conversion error, so capacitor matching is one of the primary factors limiting the linearity of a charge-redistribution SAR ADC. This is why the array is built from identical unit capacitors and laid out with careful attention to symmetry and process gradients, as discussed in Chapter 5.

2.4 Split Capacitor Array

A drawback of a purely binary-weighted array is that its total capacitance grows exponentially with resolution. From $C_{\text{tot}} = 2^N C$, a 12-bit converter would require 4096 unit capacitors in each array. Since the unit capacitor cannot be made arbitrarily small (it is bounded by matching and by kT/C noise requirements) such an array

becomes prohibitively large, with severe penalties in area, reference-driver power, and settling time [1].

To keep the capacitance manageable, a *split* (or bridged) capacitor array is used. The array is divided into a most-significant-bit (MSB) sub-array and a least-significant-bit (LSB) sub-array, connected by a single *bridge* capacitor C_b . The bridge capacitor attenuates the contribution of the LSB sub-array so that, when viewed from the MSB side, the LSB array behaves as a fractional capacitor with the correct binary weight. This allows an N -bit array to be split into two sub-arrays of roughly $N/2$ bits each, reducing the largest capacitor and the total capacitance dramatically compared with a single monolithic array [6].

For the array to preserve its binary weighting across the split, the bridge capacitor must be sized so that the series combination of C_b with the LSB sub-array presents the correct effective capacitance to the MSB node. For an array split into two M -bit halves, this condition gives a bridge capacitance of

$$C_b = \frac{2^M}{2^M - 1} C, \quad (2.5)$$

where C is the unit capacitance. The specific split used in this design, together with the resulting value of C_b and the practical consequences for layout and matching, is described in Chapter 4 and Chapter 5.

2.5 Capacitor Switching Schemes

The sequence in which the DAC capacitors are switched during a conversion has a strong influence on the energy the converter consumes, because each switching event moves charge between the reference source and the capacitor array. In the conventional switching scheme, the capacitor under test is first switched up to the reference and, if the comparator decision requires it, switched back down again on the following step. These up-and-down transitions, and in particular the large transition associated with the most significant bit, make the conventional scheme relatively energy-inefficient. A number of more efficient switching schemes have therefore been proposed to reduce this energy.

The scheme adopted in this work is the *monotonic* switching procedure introduced by Liu et al. [7]. In this scheme the input is sampled on the top plates of the capacitor array and, after each comparison, only the capacitor corresponding to the just-decided bit is switched, always in a single direction. Because no capacitor is ever switched back, each conversion involves fewer and smaller charge transfers than the conventional approach. Liu et al. report that this reduces the average switching energy by approximately 81% and the total array capacitance by about 50% compared with the conventional scheme [7]. The unidirectional nature of the

procedure also simplifies the digital control logic, since each bit requires only a single switching decision.

Monotonic switching does cause the input common-mode voltage at the comparator to drift during a conversion, which places requirements on the comparator design; this is addressed in the comparator design described in Chapter 4. Combined with a split capacitor array, monotonic switching allows a compact and energy-efficient DAC, which is why it is well suited to the low-power, differential converter targeted in this work [6, 7]. The precise arrangement used, including how the scheme maps onto the fully differential structure, is described in Chapter 4.

2.6 The Dynamic Comparator

The comparator is the only active analog block inside the conversion loop, and its behaviour sets much of what the converter can achieve. Its task is simple to state: at each step of the binary search it decides which of its two inputs is larger and produces a single-bit digital result for the SAR logic. How well it does this, and how much energy it costs, depends on the circuit used to make the decision.

Two broad options exist. A static comparator built around a pre-amplifier draws current continuously, which gives it good gain and robustness but makes it a poor fit for a low-power converter. A dynamic comparator instead makes its decision on a clock edge and draws current only during that decision, settling to a stable output with no static consumption. This is the property that makes the SAR ADC attractive in the first place, since it means the converter draws almost no power between conversions. For this reason the dynamic comparator is the standard choice in low-power SAR designs, and it is the approach followed in this work.

Three non-idealities of the comparator matter for the converter as a whole. The first is *offset*, a fixed voltage difference between the inputs at which the comparator actually switches, caused mainly by device mismatch. A static offset shifts every code transition by the same amount and so appears as an input offset of the whole converter, which is usually harmless for a dynamic input but must be bounded. The second is *noise*: the comparator adds its own thermal noise to each decision, and near a code boundary this noise can flip the result. Comparator noise adds to the sampling noise of Section 2.8 in setting the converter's overall noise floor. The third is *metastability*, the case where the two inputs are so close that the comparator cannot resolve a clear result within the time available, leaving an undefined output. In a SAR ADC this is rare but not impossible, since the comparator input becomes small near the end of a conversion.

A common way to meet the gain, speed, and noise requirements together is to precede a regenerative latch with a small dynamic pre-amplifier, so that the

latch resolves quickly while the pre-amplifier keeps its offset and noise contribution low [8]. The specific comparator topology used in this design, and the reasoning behind it, are given in Chapter 4.

2.7 Fully-Differential Operation

The converter in this work is built as a fully differential circuit, meaning the signal is carried as the difference between two complementary paths rather than as a single voltage referred to ground. The capacitor array, the comparator, and the switching all exist in mirrored pairs. This choice has a cost, since it roughly doubles the number of capacitors and switches, but it brings several advantages that are worth that cost in a converter aiming for 12-bit linearity.

The first advantage is signal range. A differential structure can represent a swing twice as large as a single-ended one for the same supply, since the two paths move in opposite directions. This larger signal raises the available dynamic range by about 6 dB, which directly eases the noise and matching requirements elsewhere in the converter.

The second advantage is rejection of common-mode disturbances. Any interference that appears equally on both paths, such as supply noise or substrate coupling, is removed when the comparator takes the difference between them. The same is true of charge injected by the switches, as long as it couples equally into both sides, which careful symmetric layout is meant to ensure.

The third advantage concerns linearity. In a symmetric differential circuit the even-order distortion terms appear identically on both paths and cancel when the difference is taken, so the dominant remaining distortion is odd-order. Suppressing the even-order terms improves the spurious-free dynamic range, which is one of the metrics defined in Section 2.9. Together these properties make the differential structure the natural choice for this design, and the way the switching scheme maps onto the two arrays is described in Chapter 4.

2.8 Noise in the Sampling Process

Beyond the deterministic errors introduced by capacitor mismatch, the capacitive DAC is also subject to a fundamental thermal-noise limit that arises during sampling. When the input is sampled onto the capacitor array through the finite on-resistance of a sampling switch, the thermal noise of that switch is sampled together with the signal. Integrating the thermal noise of the switch resistance over the bandwidth of the sampling network yields the well-known result that the mean-square sampled noise voltage depends only on temperature and capacitance, and not on the switch

resistance itself:

$$\overline{v_n^2} = \frac{kT}{C_{\text{tot}}}, \quad (2.6)$$

where k is the Boltzmann constant, T the absolute temperature, and C_{tot} the total sampling capacitance [1].

Equation (2.6) sets a lower bound on the sampling capacitance. For the quantisation noise of the converter not to be dominated by thermal noise, the sampling capacitance must be large enough that the kT/C noise remains below the least-significant-bit (LSB) level of the converter. Increasing the resolution therefore forces a larger sampling capacitance, which in turn increases area, the energy required to charge the array, and the settling time of the reference network.

This creates a three-way trade-off that governs the choice of unit capacitance. The unit capacitor must be large enough to satisfy the kT/C noise requirement and to achieve adequate matching, yet small enough to keep the total array area, power, and settling time within acceptable limits. The specific unit capacitance chosen for this design, and the resulting total array capacitance, are derived from these considerations in Chapter 4.

2.9 Performance Metrics

To judge how well a converter actually works, and to place it alongside other designs, a common set of metrics is needed. These fall into two groups: static metrics, which describe the converter's transfer characteristic under slowly changing inputs, and dynamic metrics, which describe its behaviour with a fast-moving signal. A pair of figures of merit then condense resolution, speed, and power into a single number for comparison. The definitions below are the ones used to characterise the converter in Chapter 6, and they follow standard practice in the data-converter literature [9].

2.9.1 Static Metrics

Static metrics measure how far the real transfer curve strays from the ideal staircase. The *differential nonlinearity* (DNL) is the error in the width of each code, expressed in units of one least-significant bit (LSB):

$$\text{DNL}(k) = \frac{V(k+1) - V(k)}{V_{\text{LSB}}} - 1, \quad (2.7)$$

where $V(k)$ is the input voltage at the k -th code transition and $V_{\text{LSB}} = V_{\text{FS}}/2^N$ is the ideal step size for an N -bit converter of full-scale range V_{FS} . If every code satisfies $\text{DNL}(k) > -1 \text{ LSB}$, no code is ever skipped and the converter is said to have *no missing codes*. The *integral nonlinearity* (INL) accumulates these local errors into the overall deviation from a straight line, and is obtained by summing the DNL:

$$\text{INL}(k) = \sum_{i=0}^k \text{DNL}(i) \quad (2.8)$$

A converter is *monotonic* when its output never decreases as the input rises; keeping the DNL above -1 LSB everywhere is enough to guarantee this. As noted in Section 2.5, the monotonic switching scheme used here supports this property by construction.

2.9.2 Dynamic Metrics

Dynamic metrics are read from the output spectrum when a pure sine wave is applied at the input, and they capture how the converter behaves under realistic, fast-changing signals. The *signal-to-noise ratio* (SNR) compares the signal power with the total noise power, leaving harmonics aside, while the *signal-to-noise-and-distortion ratio* (SNDR) folds the harmonic distortion back in:

$$\text{SNDR} = 10 \log_{10} \frac{P_{\text{signal}}}{P_{\text{noise}} + P_{\text{distortion}}} \quad (2.9)$$

The *spurious-free dynamic range* (SFDR) is the distance from the fundamental to the largest single spur in the spectrum, and the *total harmonic distortion* (THD) is the ratio of the combined harmonic power to the fundamental. The single most useful summary of dynamic performance is the *effective number of bits* (ENOB), which expresses the measured SNDR as the resolution an ideal quantiser would need to achieve that same SNDR:

$$\text{ENOB} = \frac{\text{SNDR} - 1.76}{6.02}, \quad (2.10)$$

with SNDR in decibels. It comes from the ideal quantisation-noise limit of Equation (2.3). The gap between the nominal resolution N and the ENOB is therefore a direct measure of how much real-world noise and distortion cost the converter.

2.9.3 Figures of Merit

Comparing converters that differ in resolution, speed, and power calls for a figure of merit (FOM) that rolls these quantities into one number. The most widely quoted is the *Walden FOM*, which normalises the power to the sampling rate and to the effective resolution [10]:

$$\text{FOM}_W = \frac{P}{2^{\text{ENOB}} \cdot f_s}, \quad (2.11)$$

where P is the total power and f_s the sampling frequency; the result is an energy per conversion step, usually quoted in femtojoules. A smaller Walden FOM means a more energy-efficient converter, which is why it is the headline figure for low-power SAR designs [11]. Because it charges only a factor of two for each extra effective bit, however, it tends to flatter lower-resolution converters and penalise high-resolution

ones. The *Schreier FOM* offers the complementary view, referencing the achieved dynamic range to the Nyquist bandwidth [12]:

$$\text{FOM}_S = \text{DR} + 10 \log_{10} \left(\frac{f_s/2}{P} \right), \quad (2.12)$$

where the dynamic range DR, in practice approximated by the peak SNDR, is in decibels and the result is likewise in decibels. Here the convention flips: a *larger* Schreier FOM is better. Reporting both figures gives a fairer picture of a converter that has to balance resolution against power, and both are quoted for the proposed design in Chapter 6.

Chapter 3

Related Work

This chapter reviews the work that this thesis builds on. It first covers the open-source process design kits and EDA tools that make a fully open converter design possible, and then the SAR ADC techniques the design relies on: the switching schemes, the array structures, and the comparator. The chapter ends with a comparison against recent implementations to show where this work stands.

3.1 Open-Source PDKs and EDA Flows

For most of the history of chip design, the process design kit has been proprietary and available only under non-disclosure. This has been a problem because a published design cannot be reproduced or independently checked if the models and rules behind it are not available. Open-source PDKs, which combine a manufacturable process with models and rules that anyone can download, have started to change this.

The SkyWater SKY130 process, released through a collaboration between Google and SkyWater, was the first to be widely adopted, and it has been used for designs ranging from standard-cell characterisation to full mixed-signal blocks [13]. The IHP SG13G2 process was released more recently. It is a 130 nm SiGe BiCMOS technology and the first open-source PDK developed in Europe, and it is the process used in this work [2, 14]. Unlike a purely digital open flow, these kits provide the analog primitives a data converter needs. The open-source tool flow has developed alongside the PDKs. Schematic capture, simulation, layout, and physical verification are all available as free tools, and a digital block can be taken from register-transfer level to layout using an automated flow that has been proven in silicon [15].

The closest work to this thesis is the open-source SAR ADC by Fath, Moser, and colleagues, a self-clocked 12-bit non-binary fully differential converter designed entirely with open-source software on SKY130 [16]. It shows that a complete mixed-

signal converter, including its analog cells, can be hardened through an otherwise digital flow. This work follows the same open-source approach but uses a different process, and its circuit-level choices differ. So far only a few converters have been demonstrated on a fully open flow, and almost all of these use SKY130. This is the gap the present work addresses.

3.2 SAR ADC Techniques

The energy a SAR ADC consumes is largely determined by how its capacitive DAC is built and switched, and both have been improved steadily over the years. Tang et al. report that ADC energy efficiency improved by roughly 7500 times in the two decades up to 2019, with about a factor of 100 coming from technology scaling and the remaining factor of 75 from architectural and circuit-level improvements [17]. The SAR architecture has benefited the most from this, which is one reason it is now the common choice for low-power conversion [18]. The underlying mechanisms were described in Chapter 2; here the aim is to show how the published approaches relate to each other and to the choices made in this work.

The conventional switching scheme is the baseline for comparison. A capacitor is switched to the reference and then switched back if the next decision requires it, and these reversals, especially on the MSB, consume energy. Liu et al. reduced this cost with the monotonic scheme, in which the input is sampled on the top plates and each capacitor is switched only once and in one direction. Compared with the conventional scheme, they report about 81 % lower switching energy and around half the array capacitance [7]. The exact saving depends on how the comparison is set up: benchmarked at equal capacitance, Tang et al. report the monotonic saving as 63 % [17]. This is the same scheme measured against a different baseline, not a contradiction. The drawback of monotonic switching is that the common-mode voltage falls throughout the conversion, which affects the comparator's offset and noise.

Two other schemes avoid this by handling the MSB differently. The V_{cm} -based scheme of Zhu et al. inserts a phase after sampling in which all bottom plates are switched to V_{cm} , so the MSB is resolved without any capacitor switching and independently of capacitor mismatch; this halves the array and moves the worst-case DNL away from mid-scale, at the cost of n additional switches to perform the V_{cm} reset [19]. The merged capacitor switching scheme of Hariprasath et al. achieves a similar result by a different arrangement [20]. Neither scheme is used in this work, but together they show the trade-off that the monotonic scheme avoids: less common-mode drift, but at the cost of an added reference and more switching logic.

Array size is a separate issue from switching. A binary-weighted array doubles in size with each additional bit, so for higher resolutions it is split into sub-arrays joined by a bridge capacitor, which reduces both the largest capacitor and the total capacitance [6]. A related idea is to give up strict binary weighting: Kuttner introduced a non-binary SAR in which the capacitor weights overlap, so that a wrong early decision can still be corrected later in the conversion [21]. Split arrays are common in higher-resolution SAR converters and are often combined with this kind of redundancy, as in the 12-bit designs of Li et al. and Kuo et al. [6, 22]. The redundancy recovers some of the settling and matching margin lost to the split, at the cost of a few extra conversion steps.

The comparator is the third block that has received attention. In a low-power SAR it is the only active element that draws no static current, but it still sets much of the noise floor [17]. The dynamic StrongARM latch, which draws current only when clocked, is the usual starting point, and its offset and noise behaviour are well documented [23]. The sampling front end is also important: a bootstrapped switch keeps the sampling device's gate-source voltage constant so that its on-resistance does not vary with the input, which preserves linearity at higher resolutions [24]. This work combines these elements, using monotonic switching on a split array, resolved by a dynamic comparator, in a fully differential structure. The reasons for these choices are given in Chapter 4.

3.3 Comparison of Recent SAR ADC Implementations

Table 3.1 compares a set of SAR ADCs with the converter developed in this thesis. The comparison is limited to designs of similar resolution and covers two groups: converters fabricated in mature commercial processes, which report silicon-measured results, and the open-source converter closest to the approach used here. The aim is not to claim that this work competes on raw performance with silicon-proven commercial designs, but to show where an open-source converter sits relative to them and what is realistic on a fully open flow.

A few points follow from the table. The commercial designs, such as the 100-MS/s converter of Li et al. [6] and the 1-MS/s converter of Kuo et al. [22], reach figures of merit in the single-digit to low-tens femtojoule range, which reflects both a mature process and the benefit of silicon-measured optimisation. These set the performance level that a proprietary flow makes possible. Their resolutions and sampling rates bracket the operating point of this work, which confirms that a 12-bit, low-megahertz converter is a well-established design point.

The more relevant comparison is with the open-source converter of Fath, Moser et al. [16]. It shares a resolution, a similar sampling rate, and an open-source design

Table 3.1 – Comparison with other SAR ADC implementations.

	This work	[16]	[22]	[6]	[25]
Year	2026	2023	2021	2021	2010
Technology (nm)	130	130	180	65	65
Process	SG13G2	SKY130	CMOS	CMOS	CMOS
Open-source PDK	Yes	Yes	No	No	No
Resolution (bit)	12	12	12	12	10
Sample rate (MS/s)	2	1.44	1	100	1
Supply (V)	1.2	1.8	0.95/1.4 ^c	1.2	1.0
ENOB (bit)	11.3 ^a	— ^b	11.25	10.17 ^d	8.75
SNDR (dB)	69.5 ^a	— ^b	69.51	—	54.4
Power (μ W)	272 ^a	703	—	800	1.9
FOM _W (fJ/conv.)	54 ^a	— ^b	18.39	6.94 ^d	4.42
Area (mm ²)	— ^e	0.175	0.367	0.029	—
Result type	Pre-layout	Post-layout	Measured	Measured	Measured

^aPre-layout simulation, typical corner, 27 °C^bDynamic metrics not reported in [16].^cSeparate supplies for digital (0.95 V) and analog (1.4 V).^dLow-frequency input; at Nyquist (50 MHz) the same design reports ENOB 8.83 and FOM_W 17.6 fJ.^eDelivered as a macro; a directly comparable full-chip area is not defined.

flow with this work, which makes it the natural reference point. The two differ in process (SKY130 versus SG13G2), coding (non-binary versus binary), and timing (self-clocked versus synchronous), so the comparison shows not a like-for-like circuit but what can be achieved on an open flow with different design choices. Providing a second open-source result, in a different open PDK, is useful in itself, because it shows that the approach is not limited to one technology.

The biomedical converter of Naveed et al. [26] represents the low-power, low-bandwidth end of this design space, and it is typical of the sensor and biomedical applications this class of converter is intended for.

Overall, the table places this work in a modest but meaningful position: a 12-bit differential SAR ADC with target metrics in line with the established design space, realised entirely on an open flow.

3.4 Positioning of This Work

The work reviewed in this chapter frames the contribution of this thesis. The techniques the design is based on, namely monotonic switching, the split capacitor array, the dynamic comparator, and the bootstrapped sampling switch, are all well established, and no claim to novelty is made for any single block. Similarly, 12-bit SAR converters in the low-megahertz range are a mature and well-populated design point, as the comparison in Section 3.3 shows.

What distinguishes this work is the combination of a complete, fully differential 12-bit converter with a fully open-source design and verification flow, implemented in the IHP SG13G2 BiCMOS process. Very few converters have so far been demonstrated

on a fully open flow, and those that have are mostly on SKY130. This thesis adds an independent result on a different open PDK, which is useful because it tests whether the open-source approach carries over to another technology. The contribution is therefore less a performance record than a reproducible, openly documented design that others can inspect, rebuild, and extend, which is difficult with converters designed under proprietary kits.

It is worth stating clearly what this work does and does not claim. It does not aim to match the figures of merit of silicon-proven commercial converters, and it does not introduce a new circuit technique. It aims to show that a 12-bit differential SAR ADC can be designed, laid out, and verified end to end with open tools on an open PDK, and to document that process in enough detail to be reproduced. The remaining chapters describe how this was done: Chapter 4 develops the architecture and the reasoning behind the design choices, Chapter 5 covers the circuit and physical implementation, and Chapter 6 presents the results.

Chapter 4

Architecture and Concept

Chapter 2 introduced the building blocks of a SAR ADC in general terms and deferred every specific design choice to this chapter. Here those choices are made and described. The chapter follows the signal path through the converter, one block at a time, and for each block explains what was chosen and why, referring forward to Chapter 5 for the circuit- and layout-level detail and to Chapter 6 for the results that the choices produce.

4.1 Architecture Overview

The converter is a fully differential, synchronous 12-bit SAR ADC using monotonic switching. Its top-level structure is shown in Figure 4.1 and follows directly from the operating principle of Section 2.1, with each generic block now given a concrete implementation.

The input is taken by a bootstrapped sampling switch, which samples the differential signal onto the top plates of the capacitive DAC. The DAC is a split binary-weighted array of metal–insulator–metal (MIM) capacitors, divided into a 6-bit most-significant sub-array and a 6-bit least-significant sub-array joined by a bridge capacitor. During conversion the array is switched according to the monotonic scheme, the bottom plates being driven between V_{ref} and ground by a switch array on each side under control of the SAR logic, so that the top-plate voltages form the residues compared at each step. The comparison is made by a dynamic comparator consisting of a static preamplifier followed by a StrongARM latch. The preamplifier provides gain and isolates the latch from the DAC, while the latch resolves each bit decision. The comparator output drives the synchronous SAR logic, which registers each decision, updates the switch array for the next step, and, after twelve steps, presents the 12-bit output code.

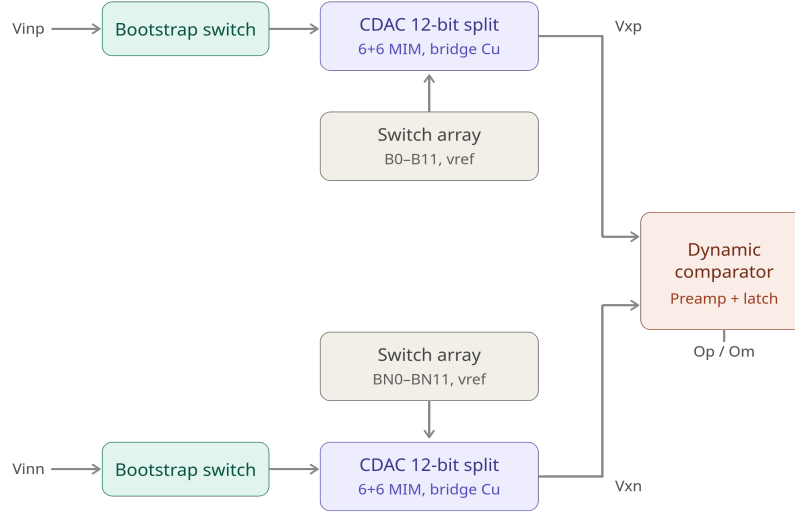


Figure 4.1 – Top-level architecture of the fully-differential 12 bit SAR ADC. The differential input is sampled by bootstrapped switches onto two split binary-weighted CDACs, whose bottom plates are driven by switch arrays under control of the SAR logic. The resulting top-plate voltages V_{xp} and V_{xn} are compared by a dynamic comparator consisting of a preamplifier followed by a StrongARM latch.

The remainder of the chapter treats these blocks in the order they are designed: the capacitive DAC and its switching, including the switch array (Sections 4.2 and 4.3), the comparator (Section 4.4), the sampling switch (Section 4.5), and the SAR logic (Section 4.6). The choice of a fully differential structure, motivated in Section 2.7, underlies all of them: every analog block except the comparator exists as a mirrored pair, and the switching scheme is applied symmetrically to the two halves of the array.

4.2 Capacitive DAC

The DAC is the core of the converter. It samples the input, holds it, and generates the reference levels for the binary search, and its accuracy sets the linearity of the whole converter, so its design receives the most attention here.

4.2.1 Split Array and Unit Capacitor

A monolithic 12-bit binary-weighted array would require $2^{12} = 4096$ unit capacitors per side, which, as shown in Section 2.4, is impractical in area, reference-driver

power, and settling time. The array is therefore split into a 6-bit MSB sub-array and a 6-bit LSB sub-array joined by a bridge capacitor, so that each sub-array contains only $2^6 = 64$ unit capacitors. This reduces the largest capacitor and the total array capacitance by roughly two orders of magnitude relative to the monolithic case, while preserving the 12-bit weighting.

The unit capacitor is a metal–insulator–metal (MIM) capacitor. Its value is bounded from below by two requirements, the sampled thermal noise and the accuracy of the capacitor weighting, and from above by area, switching energy, and settling time.

Sampling-noise bound

The 12-bit code resolves the magnitude of the differential input over a range of $V_{FS} = 1.2\text{ V}$, set by the reference; the polarity is handled separately by a sign bit, as described in Section 4.6. The least-significant-bit step is therefore

$$V_{LSB} = \frac{V_{FS}}{2^{12}} = \frac{1.2\text{ V}}{4096} \approx 293\text{ }\mu\text{V}. \quad (4.1)$$

Because the converter is differential, the input is sampled onto two capacitor arrays, each contributing an independent kT/C noise term. The two are uncorrelated, so their variances add in the differential signal:

$$\sigma_{\text{diff}}^2 = \frac{2kT}{C_{\text{tot}}}. \quad (4.2)$$

Requiring the sampling noise to stay within half an LSB with a 3σ margin gives a target of

$$\sigma_{\text{diff}} < \frac{V_{LSB}}{6} = \frac{293\text{ }\mu\text{V}}{6} \approx 48.8\text{ }\mu\text{V}, \quad (4.3)$$

which bounds the total array capacitance per side as

$$C_{\text{tot}} \geq \frac{2kT}{\sigma_{\text{diff}}^2} = \frac{2(1.38 \times 10^{-23}\text{ J/K})(300\text{ K})}{(48.8\text{ }\mu\text{V})^2} \approx 3.48\text{ pF}, \quad (4.4)$$

and, with 64 unit capacitors per side, a minimum unit capacitance of

$$C_{\text{u,min}} = \frac{3.48\text{ pF}}{64} \approx 54.4\text{ fF}. \quad (4.5)$$

Weighting accuracy and the chosen value

Equation (4.5) is a noise bound only, and in this design it is not the binding constraint. A unit capacitor sized at that limit was found in simulation not to reproduce the correct output code across the input range: the converter’s accuracy is set not only by noise but by how faithfully the array realises its intended binary weights, and at small unit values the parasitic capacitance of the array nodes and the routing becomes a significant fraction of the unit element itself, corrupting the weighting.

Random mismatch between nominally identical MIM capacitors acts in the same direction, its relative standard deviation falling as the capacitor area increases [27]. Both effects therefore push the unit capacitor above the noise-derived minimum.

The unit capacitor was accordingly sized by simulation, starting from $C_{u,\min}$ and increasing it until the converter reproduced the correct output code across the full input range. Values of two and two-and-a-half times $C_{u,\min}$ did not, while three times did. The unit capacitor is therefore 165.2 fF, realised as a square MIM element of side $10.44\text{ }\mu\text{m}$, an area of $109\text{ }\mu\text{m}^2$, which at the SG13G2 specific capacitance of $1.5\text{ fF}/\mu\text{m}^2$ together with the perimeter contribution of $40\text{ aF}/\mu\text{m}$ gives

$$C_u = C'A + C'_p P \approx 165.2\text{ fF}. \quad (4.6)$$

The total array capacitance is then $C_{\text{tot}} \approx 10.5\text{ pF}$ per side, and the resulting differential sampling noise is

$$\sigma_{\text{diff}} = \sqrt{\frac{2kT}{C_{\text{tot}}}} \approx 28.1\text{ }\mu\text{V} \approx 0.10\text{ }V_{\text{LSB}}, \quad (4.7)$$

so the 3σ sampling noise stays within 0.30 LSB, well inside the half-LSB budget of Equation (4.3). The array is thus limited by weighting accuracy rather than by noise, with the noise requirement exceeded by a wide margin.

Terminating dummy capacitor

The LSB sub-array also contains a terminating dummy capacitor equal to one unit capacitor, $C_{\text{dummy}} = C_u$, whose bottom plate is permanently tied to V_{ref} and is never switched. This dummy element completes the binary weighting so that the total array capacitance follows $C_{\text{tot}} = 2^N C_u$, as derived in Section 2.4. It is a separate element from the bridge capacitor of Section 4.2.2: the bridge joins the MSB and LSB sub-arrays, whereas the dummy terminates the LSB array.

4.2.2 Bridge Capacitor

For an exact 6+6 split, the bridge capacitor must scale the LSB sub-array so that it presents the correct fractional weight to the MSB node. From Equation (2.5), the ideal value for two 6-bit halves is

$$C_b = \frac{2^6}{2^6 - 1} C = \frac{64}{63} C \approx 1.016 C, \quad (4.8)$$

where C is the unit capacitance. Rather than realise this non-integer value, this design uses a plain unit capacitor as the bridge, $C_b = C$. The motivation is layout regularity and matching: a bridge built from the same unit element as the rest of the array can be placed and surrounded identically to its neighbours, avoiding the irregular geometry and the associated parasitic and matching penalties that a custom $1.016C$ element would introduce. The cost is a small, bounded weighting error,

since the LSB sub-array is now attenuated by a slightly incorrect factor. This error is deterministic rather than random, affects only the relative weight of the lower six bits, and is reflected in the static linearity reported in Chapter 6. The trade-off is therefore a deliberate exchange of a small, characterised systematic error for a more regular and better-matched array.

4.3 Switching Scheme

The DAC of Section 4.2 is operated with the monotonic switching procedure of Liu et al. [7]. The scheme was introduced in general terms in Section 2.5; this section describes how it is applied to the fully differential array of this design and why it suits the converter.

At the start of a conversion the input is sampled onto the top plates of the two capacitor arrays through the bootstrapped sampling switch, while every bottom plate is held at the reference voltage V_{ref} . Because the signal is taken on the top plates, the comparator can make the first bit decision immediately, with no capacitor switching. This is the origin of one of the scheme's two advantages: for an N -bit converter the array needs only 2^{N-1} unit capacitors per side rather than 2^N , since the most significant decision costs no switching [7].

After the first comparison, the SAR logic switches a single capacitor. On the side whose top plate sits at the higher potential, the corresponding unit capacitor is switched from V_{ref} down to ground, while the capacitor on the opposite side is left unchanged. This pulls the higher side down toward the lower one by the weight of the switched capacitor. The comparator then makes the next decision, and the procedure repeats: at every step exactly one capacitor is switched, always from V_{ref} to ground, and always on the higher-potential side. Because each capacitor moves only once and only in a single direction, no charge is ever returned to the reference, which is why the monotonic scheme draws substantially less switching energy than the conventional trial-and-error approach [7]. The unidirectional switching also simplifies the digital control, since each bit requires only one switching decision rather than a switch-and-possibly-reverse sequence.

Figure 4.2 shows the arrangement on the differential array. The symmetry of the two sides is important: the same procedure is applied to both, and the differential output is taken as the difference between the two top-plate voltages, so the switching is inherently balanced.

A consequence of switching only downward is that the common-mode voltage at the comparator inputs does not stay constant. As capacitors are switched to ground, both top-plate voltages fall, and the input common-mode drifts steadily downward from its sampled level toward ground over the course of the conversion,

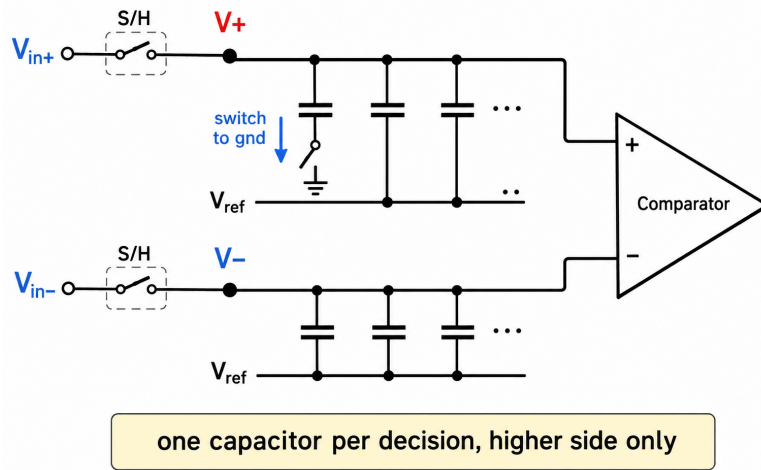


Figure 4.2 – Simplified differential capacitor array with monotonic switching. The input is sampled on the top plates while all bottom plates are held at V_{ref} , and the first comparison is made without switching. After each decision, the unit capacitor on the higher-potential side is switched to ground while the opposite side is left unchanged, giving one switching event per bit.

as illustrated in Figure 4.3. This drift is the price paid for the scheme's energy efficiency and simplicity, and it places a specific requirement on the comparator, which must continue to operate correctly as its input common-mode falls. How the comparator meets that requirement is the subject of the next section. Tang et al. note this common-mode variation as the main drawback of the monotonic scheme and discuss its effect on comparator behaviour [17].

4.3.1 Switch Array

The monotonic scheme requires each capacitor bottom plate to be connectable to either V_{ref} or ground, and to be switched between them cleanly and in step with the SAR decision. This is the function of the switch array, one per differential side, sitting between the SAR logic and the capacitor bottom plates.

Each bottom plate is driven by a transmission gate rather than a single transistor. A single NMOS passes a strong logic low but a degraded high, and a single PMOS the reverse; a transmission gate places the two in parallel, so that both V_{ref} and ground are passed to the bottom plate at their full level with a low, roughly constant on-resistance across the range. This matters because any residual voltage error on a switched bottom plate translates directly into a weighting error in the DAC. The complementary gate signals that a transmission gate requires are generated locally by inverters, which also buffer the SAR logic outputs so that each switch presents a

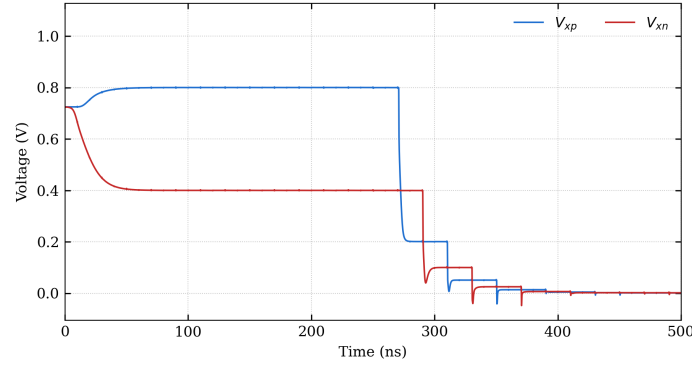


Figure 4.3 – Simulated top-plate voltages V_{xp} and V_{xn} at the comparator inputs during a single conversion cycle. After the sampling phase, the monotonic switching scheme discharges one capacitor per step on the higher-potential side only, so the two nodes step toward each other and the input common-mode falls toward ground over the twelve bit trials.

defined load to the digital block. The device sizing and layout of the array are given in Chapter 5.

4.4 Comparator

The comparator makes every bit decision, and in this design it must do so while its input common-mode voltage falls throughout the conversion, as described in Section 4.3. The comparator is based on the dynamic comparator of Liu et al. [7], preceded by a static preamplifier that was added in this work. Its structure is shown in Figure 4.4: a resistively-loaded differential preamplifier drives a StrongARM latch, which resolves the decision.

4.4.1 Input Pair and the Falling Common-Mode

The input pair of both the preamplifier and the latch is realised with PMOS devices. This follows directly from the switching scheme. Because the monotonic procedure drives the input common-mode downward, from its sampled level toward ground, the comparator must keep its input devices in saturation across that whole downward excursion. A PMOS input pair is the natural choice for an input common-mode that sits in the lower part of the supply range and moves toward ground: the devices continue to conduct and remain saturated as the common-mode falls, whereas an NMOS pair would be pushed toward and eventually out of saturation as the inputs approach ground. Liu et al. adopt a PMOS input pair for exactly this reason, noting that the comparator must function over an input common-mode range from half V_{ref}

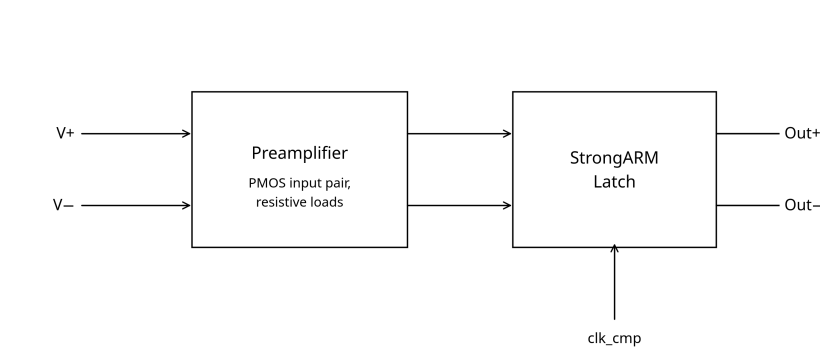


Figure 4.4 – Comparator used in this design: a static resistively-loaded PMOS-input preamplifier (left) followed by a StrongARM latch (right). The preamplifier provides gain ahead of the latch, and the PMOS input pair accommodates the falling input common-mode produced by the monotonic switching scheme.

down to ground [7]. The choice of comparator and switching scheme are therefore matched rather than independent.

4.4.2 Preamplifier

The dynamic comparator of Liu et al. resolves each decision directly with a clocked latch. In this design a static resistively-loaded preamplifier is placed ahead of that latch. Its purpose is to provide gain before the latch, so that the offset and noise contributed by the latch, referred back to the converter input, are reduced by the preamplifier gain [8]. Resistive loads are used rather than active loads because a resistor introduces no threshold-voltage mismatch of its own and keeps the load linear, which helps the matching of the two sides [8]. The load resistors are 11.7 k Ω . Their layout, including the interdigitated arrangement used to improve matching, is described in Chapter 5.

Adding a static preamplifier involves a deliberate trade-off. The dynamic comparator of Liu et al. draws no static current, which is one of the reasons the monotonic SAR is attractive for low power; the static preamplifier used here does draw a continuous current, and so reintroduces a static power component that the purely dynamic comparator avoids. This cost is accepted in exchange for the gain the preamplifier provides ahead of the latch, which lowers the input-referred offset and noise of the comparator. The contribution of the preamplifier to the total power is reported in Chapter 6.

4.4.3 Offset and Noise

Two non-idealities of the comparator matter for the converter, as introduced in Section 2.6. The static input-referred offset shifts every code transition by the same amount and appears as an offset of the whole converter; it is bounded by device sizing and by the gain of the preamplifier. A Monte Carlo analysis gives a standard deviation of the input-referred offset of approximately 3.9 LSB, which motivated the sizing of the input devices; the full distribution is reported in Chapter 6. Comparator noise adds to the sampling noise of Section 2.8 in setting the converter's noise floor, and in a low-power SAR the comparator is typically the dominant noise contributor of the active blocks [17]. The detailed comparator topology and its device sizing are given in Chapter 5.

4.5 Sampling Switch

The input is sampled onto the top plates of the capacitor array through a bootstrapped switch, one on each of the two differential inputs. The choice follows directly from the linearity target.

4.5.1 Why Bootstrapping

A plain MOS transistor used as a sampling switch has an on-resistance set by its gate-source overdrive. With the gate tied to a fixed supply, the source follows the input signal, so the overdrive, and therefore the on-resistance, varies with the input level. Two problems follow. The sampling time constant $R_{\text{on}}C_{\text{tot}}$ becomes signal-dependent, so different parts of the input waveform settle to different degrees within the same sampling window. And because the resistance modulates with the signal itself, the resulting error is nonlinear rather than a simple gain term, and appears as distortion in the output spectrum. At 12-bit resolution neither is acceptable.

A bootstrapped switch removes this dependence by holding the gate-source voltage of the sampling transistor constant. A capacitor is pre-charged to the supply during the off phase and then connected between the gate and the source of the sampling device when the switch turns on, so that $V_{\text{GS}} \approx V_{\text{DD}}$ regardless of the instantaneous input level [24]. The on-resistance is then approximately constant across the input range, the sampling bandwidth is uniform, and the signal-dependent distortion is suppressed. Liu et al. adopt a bootstrapped switch in their monotonic SAR for exactly this reason, noting that fixing the gate-source voltage at the supply makes the on-resistance a small constant value and so improves switch linearity [7].

4.5.2 Circuit Arrangement

The switch follows the bootstrapping topology introduced by Abo and Gray [28], which is the arrangement adopted in the monotonic SAR of Liu et al. [7] and remains the standard approach for linear sampling at low supply voltages. It operates from a single clock, the complementary phase being generated locally by an inverter.

During the *off* phase, the boost capacitor is charged to the supply: a PMOS device connects its top plate to V_{DD} while NMOS devices hold its bottom plate at ground. At the same time the gate of the sampling device is pulled to ground, holding the switch firmly off and isolating the array from the input.

On the *sampling* phase, the charging devices turn off and the bottom plate of the boost capacitor is connected to the input node while its top plate drives the gate of the sampling device. Because the capacitor holds V_{DD} , the gate is lifted to approximately one supply above the input, giving the sampling device a gate-source voltage of about V_{DD} that does not vary with the input level.

Sustaining a gate voltage above the supply rail requires care. A cross-coupled device pair, with its own pumping capacitors, generates a boosted control node that holds the charging PMOS firmly off during the sampling phase, when the capacitor's top plate rises above V_{DD} and would otherwise turn that device back on and discharge the boosted node. This arrangement is what allows the above-rail gate drive to be produced while keeping every transistor's terminal voltages within V_{DD} , which was the central concern of the original design [28] and matters equally here, where the process offers no low-threshold devices for the purpose.

4.6 SAR Logic and Timing

The SAR logic closes the conversion loop. It registers each comparator decision, drives the switch array to update the DAC for the next step, and assembles the output code once all bits are resolved.

4.6.1 Synchronous Operation

Two timing styles are common in SAR converters. An asynchronous controller generates its own internal clock from the comparator's decision, starting the next comparison as soon as the previous one completes, which removes the need for a fast external clock and lets each bit take only as long as it needs. A synchronous controller instead sequences the conversion from an external clock, giving every bit decision the same fixed time slot. The asynchronous approach is faster on average, but its timing depends on the comparator's own delay and it requires custom design of the delay and handshake circuitry.

This design uses a synchronous controller, driven by two external clocks: `clk_samp`, which controls the sampling phase, and `clk_cmp`. This strobes the comparator once per bit decision, the controller runs at an internal bit-clock of 50 MHz to resolve all twelve bits and sample within the 2 MS/s conversion window, leaving ample timing margin per bit. At the target rate of 2 MS/s the time available is comfortable, so the speed advantage of asynchronous operation is not needed here. The benefit of the synchronous choice is that the controller becomes a purely clocked digital block with no custom timing elements, which is what allows it to be described in a hardware description language and implemented through an automated digital flow, as described below.

4.6.2 Conversion Sequence and Output Coding

The controller maintains two registers, one associated with each side of the differential array, both initialised to all ones at reset. A bit index counts down from the most significant bit. On each comparator strobe, the decision determines which of the two registers has its bit at the current index cleared: if the negative side is the higher of the two, the corresponding bit of the negative-side register is cleared, and otherwise the bit of the positive-side register is cleared. The cleared bit is what drives the switch array to pull the corresponding capacitor on that side down to ground, realising the monotonic switching step of Section 4.3. The index then advances to the next bit and the process repeats until all twelve bits are resolved.

The converter resolves a differential input of either polarity, and the output is presented in *sign-magnitude* form: twelve bits give the magnitude and a separate sign bit gives the polarity. The sign is taken as the most significant bit of the negative-side register, which retains its reset value only if the positive input was the larger at the first comparison; it therefore records the polarity decided by that first comparison, which in the monotonic scheme is made before any capacitor is switched. The twelve-bit magnitude output is formed by selecting the register corresponding to the resolved polarity.

The consequence for resolution should be stated plainly. The twelve magnitude bits resolve one polarity, a range of $V_{FS} = 1.2\text{V}$, giving the LSB of Equation (4.1); the sign bit then extends the represented range to the full $\pm 1.2\text{V}$ differential span. The static and dynamic results reported in Chapter 6 are referred to the twelve-bit magnitude code.

4.6.3 Implementation Flow

Because the controller is a synchronous digital block, it is described in SystemVerilog at register-transfer level and taken to layout by an automated open-source flow, which synthesises the description to a gate-level netlist, places and routes it, and

produces a hardened macro that is then integrated with the custom analog blocks at top level. Using a digital flow for this block, in a converter that is otherwise full-custom, keeps the controller reproducible and easily modified: a change to the conversion sequence is a change to a few lines of code rather than a manual layout edit. This is one of the elements that makes the design reproducible on an open toolchain, in the spirit of the work discussed in Section 3.1. The flow itself, its configuration, the timing constraints applied, and the resulting macro are described in Chapter 5.

Chapter 5

Implementation

Chapter 4 set out the architecture and the reasoning behind each design choice. This chapter describes how those choices were realised: the transistor-level circuits and their sizing, the physical layout of the analog blocks, the automated flow used for the digital controller, and the verification performed at each level.

5.1 Design Flow and Tools

The converter was designed entirely with free and open-source software on the IHP SG13G2 open PDK [5]. Schematic entry and simulation were carried out in Xschem and ngspice, layout in KLayout, and physical verification with the DRC and LVS decks supplied with the PDK. The synchronous SAR controller was synthesised and hardened with LibreLane. Because reproducibility depends on the exact versions used, Table 5.1 records them together with the PDK revision.

Table 5.1 – Tools and versions used in this work.

Tool	Version
Xschem	V3.4.8RC
ngspice	42
KLayout	0.30.5
LibreLane	v3.0.0.dev45
Yosys	0.54
IHP SG13G2 PDK	Rev 1.2

5.2 Circuit Implementation

The complete top-level schematic is attached in Appendix B.

5.2.1 Capacitive DAC

The array follows the split structure of Section 4.2: two 6-bit sub-arrays joined by a bridge capacitor, each containing 64 unit capacitors. In the schematic of Figure 5.1 the binary weights are entered as single `cap_cmim` instances carrying a multiplicity factor, so that the MSB sub-array is described as $32C$, $16C$, $8C$, $4C$, $2C$ and C together with the bridge capacitor, and the LSB sub-array identically with a terminating capacitor in place of the bridge. Each instance therefore stands for the corresponding number of parallel unit elements, which is how the array is physically realised.

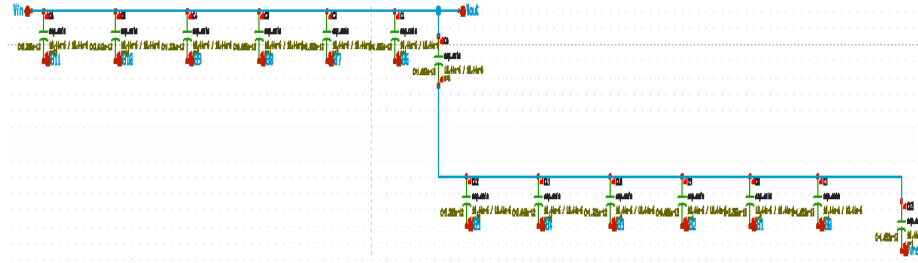


Figure 5.1 – Schematic of one capacitive DAC. The MSB sub-array (upper) and LSB sub-array (lower) are joined by the bridge capacitor. Binary weights are entered as single `cap_cmim` instances with a multiplicity factor, each representing the corresponding number of parallel unit elements.

5.2.2 Bootstrapped Sampling Switch

The sampling switch follows the topology of Abo and Gray [28], shown in Figure 5.2. Its purpose, set out in Section 4.5, is to hold the gate–source voltage of the sampling device constant at approximately V_{DD} while the switch is on, so that the on-resistance does not vary with the input level. Abo and Gray achieve this without low-threshold devices and, importantly, without subjecting any transistor to a terminal-to-terminal voltage greater than V_{DD} : although the absolute gate voltage rises above the supply during sampling, every V_{GS} , V_{GD} and V_{DS} in the circuit remains within the rated operating voltage [28].

5.2.2.1 Device sizing

The core sampling device is an NMOS of $6\text{ }\mu\text{m}$ width and $0.25\text{ }\mu\text{m}$ length, sized wide and short to keep the on-resistance low once bootstrapped. The boost capacitor C_6 is a MIM element of $11.485\text{ }\mu\text{m}$ side, giving

$$C_{\text{boost}} = C'A + C'_pP = (1.5\text{ fF}/\mu\text{m}^2)(131.9\text{ }\mu\text{m}^2) + (40\text{ aF}/\mu\text{m})(45.94\text{ }\mu\text{m}) \approx 200\text{ fF}. \quad (5.1)$$

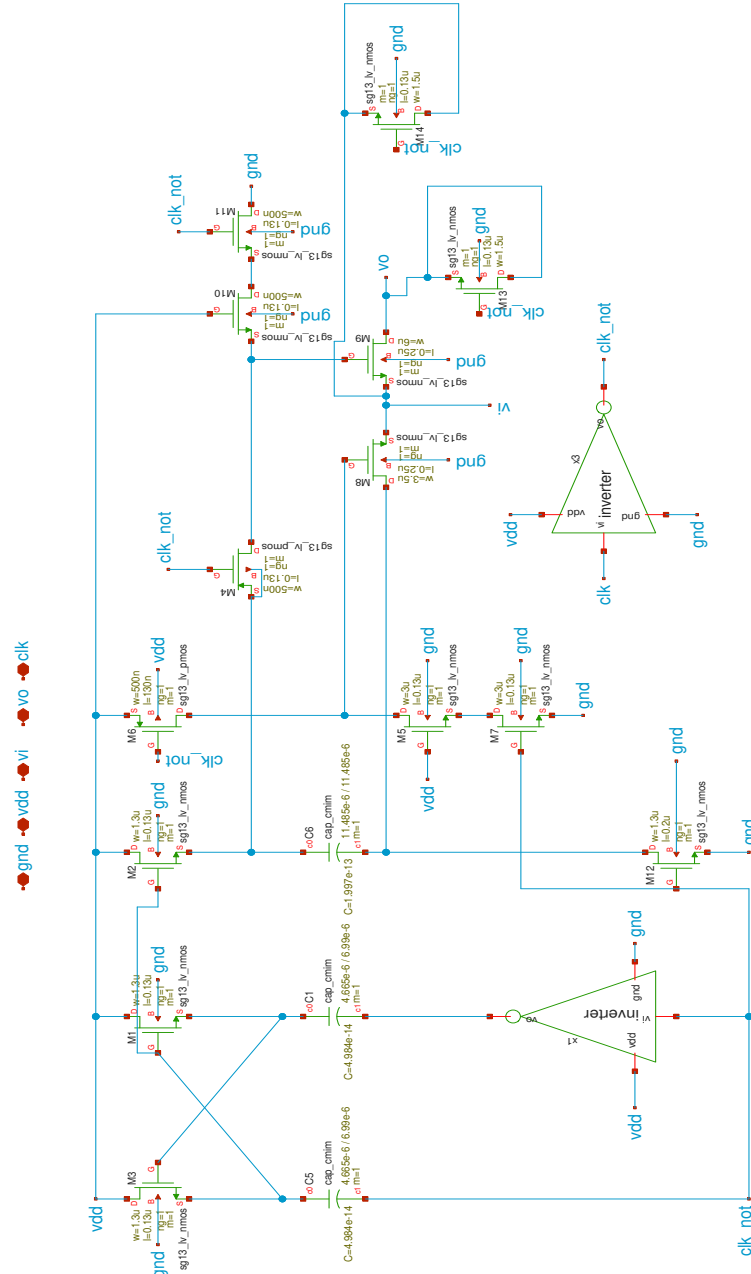


Figure 5.2 – Transistor-level schematic of the bootstrapped sampling switch, following the topology of Abo and Gray [28]. Capacitor C6 is the boost capacitor, charged to V_{DD} during the hold phase and connected between the input and the sampling gate during the sample phase. M2 is the charging device; M1, M3, C1 and C5 form the cross-coupled pair that sustains the boosted node; M9 is the sampling device.

The cross-coupled pair uses two further MIM capacitors, C1 and C5, of $4.665\text{ }\mu\text{m}$ by $6.99\text{ }\mu\text{m}$, approximately 50 fF each. The circuit runs from a single sampling clock `clk`, its complement `clk_not` being generated locally by the inverter x3.

5.2.2.2 Hold phase

During the hold phase the switch is off and the boost capacitor is charged. The charging NMOS M2, whose source is tied to V_{DD} , is turned on, connecting the top plate of C6 to the supply, while M12 on the lower branch holds its bottom plate at ground. C6 therefore charges to V_{DD} and acts, in the terminology of Abo and Gray, as the battery that will later sit across the gate and source of the sampling device [28]. At the same time the gate of the sampling device is discharged to ground, holding it firmly off and isolating the capacitor array from the input.

5.2.2.3 Sample phase

When the sampling clock goes high the charging path is broken and the bottom plate of C6 is connected instead to the input node, while its top plate is connected to the gate of the sampling device. Because C6 holds a voltage of V_{DD} across it, the gate is lifted to approximately $V_{in} + V_{DD}$, one supply above the instantaneous input. The sampling device therefore sees $V_{GS} \approx V_{DD}$ irrespective of the input level, which is the condition required for an input-independent on-resistance. As Abo and Gray note, if the source sits at V_{DD} the gate reaches $2V_{DD}$ in absolute terms, yet V_{GS} is still only V_{DD} [28]. Charge injection at the sampling instant is addressed by dummy devices of $1.5\text{ }\mu\text{m}$ width placed at the input and output nodes.

5.2.2.4 Control of the charging device

Sustaining an above-rail gate voltage requires that the charging device be held off for the whole of the sample phase. Once C6 drives the sampling gate above V_{DD} , its top plate, which is the source of M2, also rises above the supply rail; if M2 were to conduct at that moment it would discharge the boosted node back into V_{DD} and collapse the boost. A gate held at a fixed logic level would not prevent this, since the source moves above that level during sampling. The gate of M2 is therefore driven from the cross-coupled pair M1 and M3 with their pumping capacitors C1 and C5, following the clock multiplier of the original design, where the corresponding function is performed by M1, M2, C1 and C2 [28]. Because this control node is itself boosted, it tracks the top plate of C6 as it rises, so the gate–source voltage of M2 remains non-conducting throughout the sample phase and the charge on the boost capacitor is transferred unidirectionally: the capacitor is charged from the supply during the hold phase and isolated from it during sampling. The gates of M6 and

M4 are driven directly from `clk_not`: M6 turns on the NMOS M8, and M4 connects the boosted node to the gate of the sampling transistor M9.

5.2.3 Comparator

The comparator consists of the static resistively-loaded preamplifier followed by the StrongARM latch set out in Section 4.4, and is shown in Figure 5.3. Device sizes are listed in Table 5.2.

The preamplifier input pair is PMOS, for the reason given in Section 4.4, biased by a PMOS tail device whose long $0.8\text{ }\mu\text{m}$ channel raises the output impedance of the current source and improves its matching. The loads are $11.7\text{ k}\Omega$ rppd poly resistors to ground, realised as serpentine with six bends.

At the nominal bias of 0.6 V the tail supplies $27.2\text{ }\mu\text{A}$, dividing equally between the two input devices at $13.6\text{ }\mu\text{A}$ each, for a static preamplifier power of $32.7\text{ }\mu\text{W}$. The outputs rest at 159 mV , the drop across the load resistors, which drives the PMOS input pair of the latch with a gate-source overdrive of approximately 650 mV .

The simulated small-signal parameters of the input device give a preamplifier gain of

$$A_v = g_m (R_L \parallel r_{ds}) = (195\text{ }\mu\text{S})(11.4\text{ k}\Omega) \approx 2.2 \quad (7.0\text{ dB}), \quad (5.2)$$

where $r_{ds} = 493\text{ k}\Omega$ is large enough that the load resistor dominates, so the gain is set by a well-matched passive element rather than by the bias-dependent output impedance of the input devices. The input pair operates at $g_m/I_D = 14.3/\text{V}$ and remains well inside saturation, with $|V_{DS}| = 0.93\text{ V}$ against $|V_{Dsat}| = 0.15\text{ V}$.

5.2.4 Switch Array

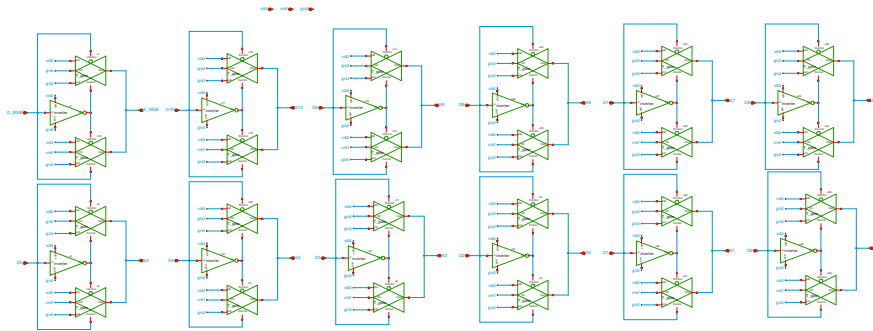


Figure 5.4 – Schematic of the switch array. Each bottom plate is driven by a transmission gate selecting between V_{ref} and ground, with local inverters generating the complementary gate signals.

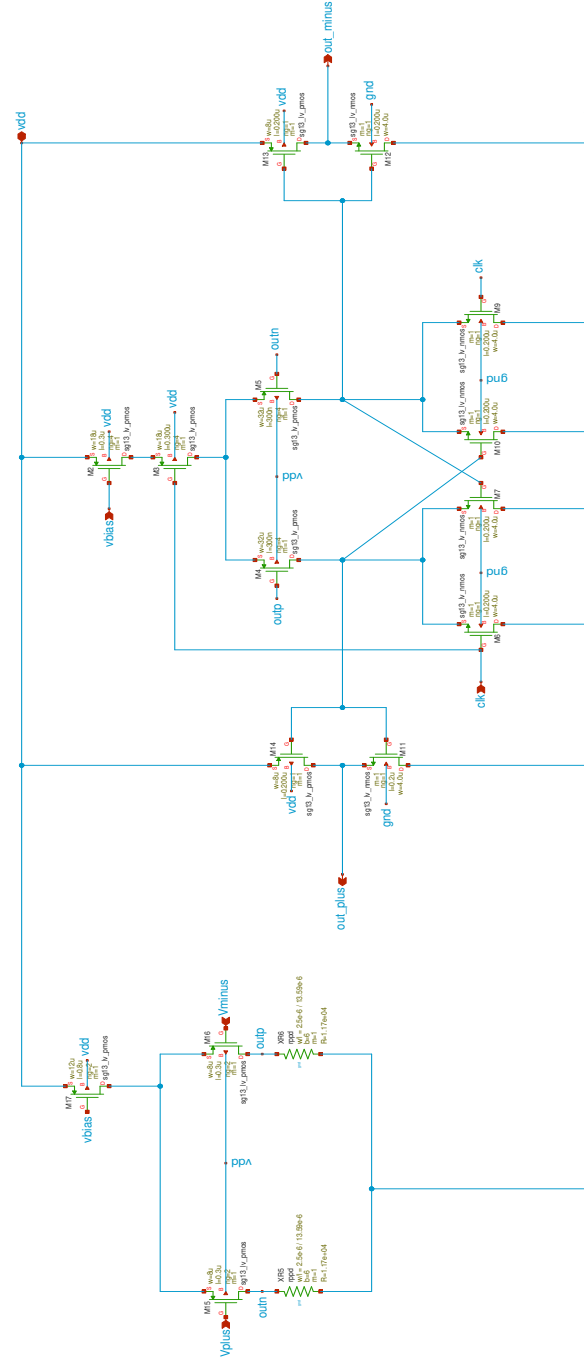


Figure 5.3 – Transistor-level schematic of the comparator: resistively-loaded PMOS-input preamplifier (left) followed by the StrongARM latch (right).

Table 5.2 – Device sizing of the comparator.

Device	Type	W (μm)	L (μm)	ng
<i>Preamplifier</i>				
M15, M16	PMOS input pair	8.0	0.30	2
M17	PMOS tail	12.0	0.80	1
R1, R2	rppd load	2.5 $\mu\text{m} \times$ 13.59 μm , 6 bends, 11.7 k Ω		
<i>StrongARM latch</i>				
M4, M5	PMOS input pair	32.0	0.30	4
M2, M3	PMOS bias/clock	18.0	0.30	4
M6–M9	NMOS latch	4.0	0.20	1
Output inverters (NMOS)		4.0	0.20	1
Output inverters (PMOS)		8.0	0.20	1

Each bottom-plate switch is a transmission gate, for the reason given in Section 4.3.1, built from an NMOS of 2 μm width and a PMOS of 4 μm width, both at the minimum channel length of 0.13 μm . The PMOS is twice the width of the NMOS to compensate for the lower hole mobility and so equalise the on-resistance of the two devices across the rail-to-rail range. The complementary gate signals are generated by local inverters with an NMOS of 0.4 μm and a PMOS of 0.8 μm width, which also buffer the SAR logic outputs driving the array.

5.3 Physical Implementation

5.3.1 Technology and Layer Stack

The SG13G2 back end of line provides five thin metal layers, Meta11 to Meta15, followed by two thick layers, TopMeta11 and TopMeta12, as shown in Figure 5.5. The MIM capacitor used throughout the converter is formed between Meta15 as the bottom plate and TopMeta11 as the top plate, with the thin nitride dielectric between them giving the specific capacitance of 1.5 fF/ μm^2 used in Section 4.2.1. The poly resistor rppd is formed in the polysilicon layer above the field oxide.

All devices are instantiated as parameterised cells from the PDK, so the internal layer usage of each capacitor, resistor and transistor follows the foundry's own construction. The layout work described below therefore concerns the placement of those cells and the routing between them. The routing strategy was to keep signal and supply connections on the lower metal layers wherever possible, using the thick upper metals sparingly, since the number of mask layers used is one of the factors that determines fabrication cost.

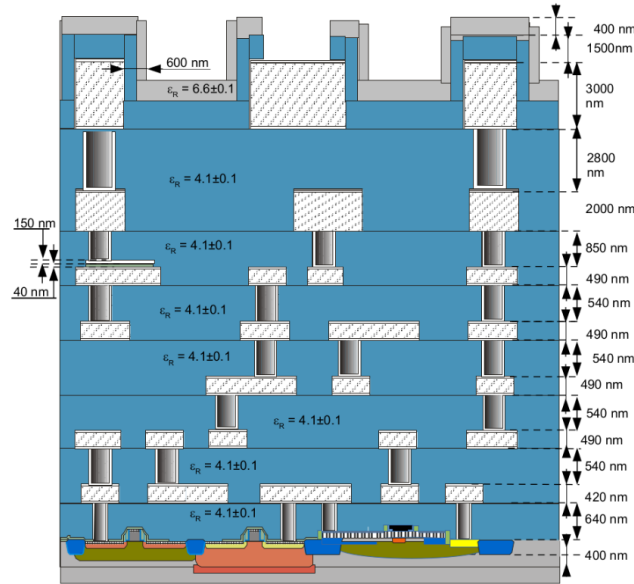


Figure 1.1.1 SG13 process cross-section (not to scale)

Figure 5.5 – Cross-section of the SG13 process stack, showing the five thin metal layers, the two thick top metals, and the dielectric thicknesses. Reproduced from the process specification [5].

5.3.2 Analog Block Layout

Figure 5.6 shows the bootstrapped switch. Its area is dominated by the three MIM capacitors: the 200 fF boost capacitor and the two 50 fF elements of the cross-coupled pair. The transistors, being small by comparison, are placed around them, with the sampling device and its dummy switches kept close to the input and output nodes to minimise the parasitic capacitance on the sampled signal path.

The switch array of Figure 5.7 is a regular arrangement of twelve identical transmission-gate cells, each with its inverter buffer, laid out in two rows of six so that the pitch matches the capacitor array it drives. The reference and ground rails run horizontally across the cell, and each switch output rises to meet the corresponding capacitor bottom plate. The regularity is deliberate: because every switch drives an identical capacitive load, keeping the cells identical keeps their delays matched, so that all bottom plates settle together.

Placement throughout was chosen to minimise the routing between blocks, keeping the interconnect parasitics on the sensitive analog nodes as small as possible. No guard rings were placed around the analog blocks. In a full chip these would normally be used to isolate the sensitive analog circuits from substrate noise gen-

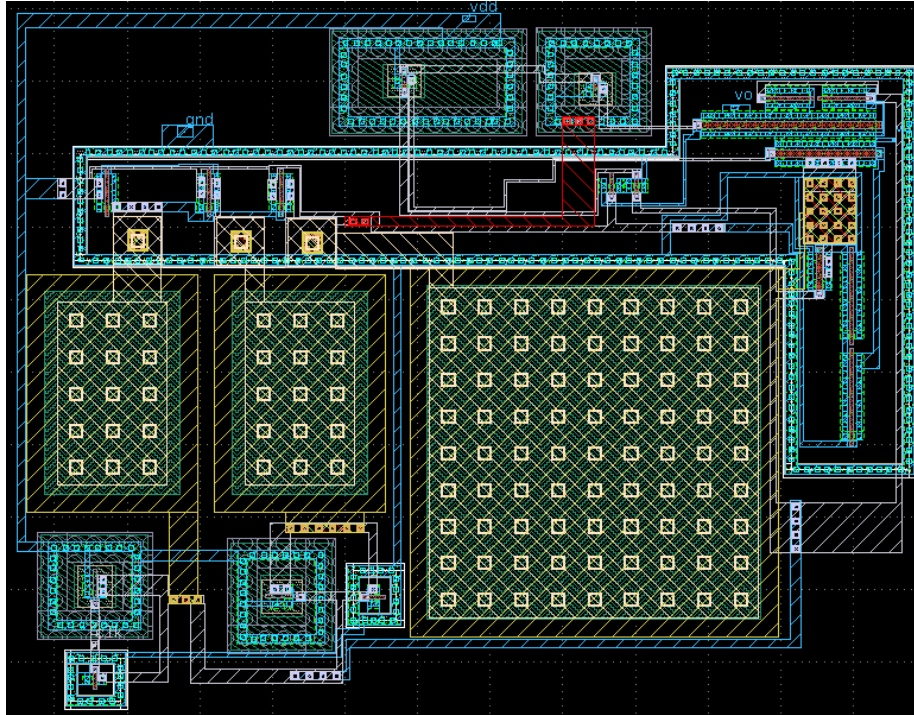


Figure 5.6 – Layout of the bootstrapped sampling switch. The three MIM capacitors are visible as the large regular arrays; the boost capacitor is the largest, at the right.

erated by the digital logic, and their absence here is a consequence of the design being delivered as a macro rather than as a complete integrated circuit.

The comparator layout is shown in Figure 5.8. The two load resistors dominate the cell area: each $11.7\text{ k}\Omega$ element is a $2.5\text{ }\mu\text{m}$ wide rppd strip of approximately $112\text{ }\mu\text{m}$ total length, folded into six segments to keep the aspect ratio practical. The two are drawn identically and placed adjacent so that they experience the same local process conditions, since it is the ratio of the two loads rather than their absolute value that sets the balance of the preamplifier. The two halves of the differential circuit are laid out symmetrically throughout for the same reason.

5.3.3 Capacitor Array Layout

Each sub-array is laid out as an 8×8 matrix of identical unit capacitors, shown in Figure 5.9, so that every weight is formed from the same physical element and the binary ratios depend only on how many elements are connected in parallel rather than on any scaled geometry.

The assignment of positions to weights is chosen to reduce the effect of process gradients across the array. The two largest weights, which dominate the linearity,

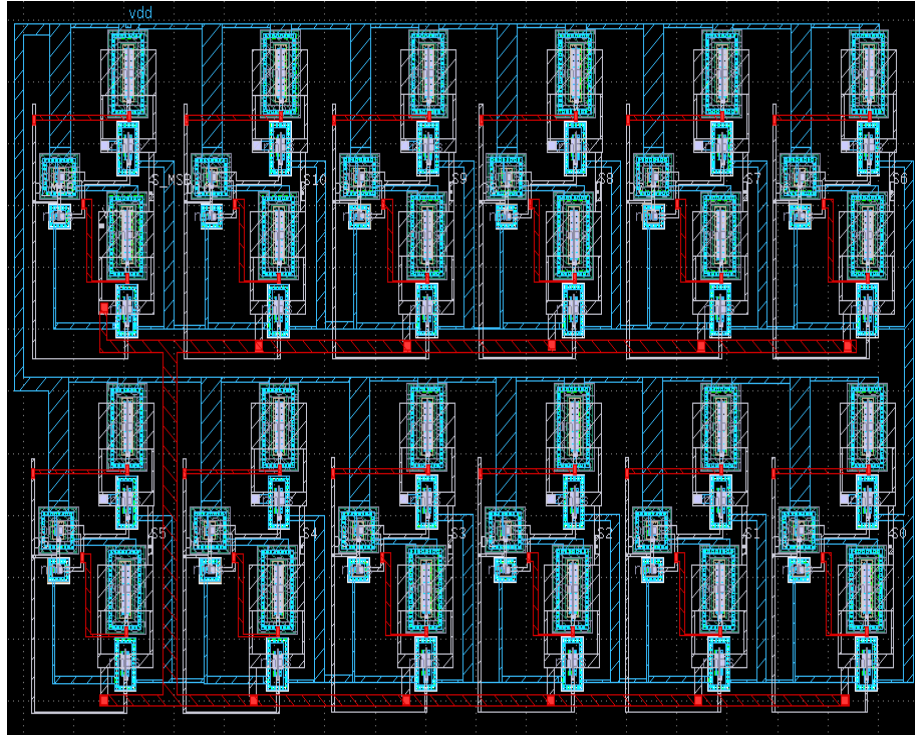


Figure 5.7 – Layout of one switch array. The twelve transmission-gate cells and their inverter buffers are arranged in two rows, with the reference and ground rails running horizontally.

are placed symmetrically about the horizontal centreline of the matrix: the 32 unit capacitors of the most significant bit occupy rows 1, 2, 6 and 7, and the 16 capacitors of the next bit occupy rows 3 and 5. A linear gradient across the array therefore affects each of these weights symmetrically and cancels to first order. The 8 capacitors of the third bit occupy the central row 4. The remaining weights, four, two and one unit capacitors, together with the bridge capacitor, occupy the final row. These smaller weights are not centroided; their contribution to the total array capacitance is small enough that the residual gradient error remains well below one LSB, and centroiding them would complicate the routing considerably for little benefit.

The bottom plate of every capacitor connects to the switch array, which selects between V_{ref} and ground according to the SAR decision, while the top plates are joined to the common node that feeds the comparator.

5.3.4 Digital Controller

The SAR controller is described in SystemVerilog and implemented with LibreLane v3.0.0.dev45 using the `sg13g2_stdcell` standard-cell library, following the flow of Figure 5.10. The design is constrained with a 10 ns clock period, correspond-

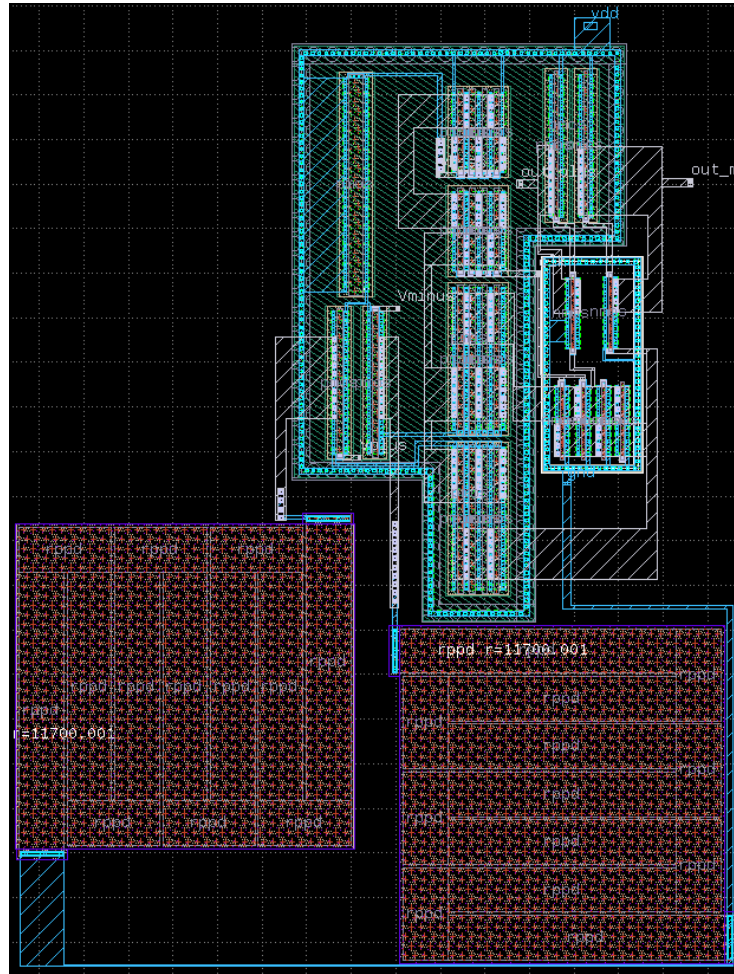


Figure 5.8 – Layout of the comparator. The two $11.7\text{ k}\Omega$ rppd load resistors occupy the lower part of the cell, drawn as matched folded serpentine.

ing to 100 MHz. The controller is clocked in the converter at 50 MHz, one edge per bit decision, so the 100 MHz synthesis constraint leaves roughly a factor of two in timing margin and ensures the block is not the speed-limiting element of the conversion.

Synthesis with Yosys 0.54 maps the description to 137 cells, of which 28 are resettable D flip-flops (sg13g2_dfrbpq_1) holding the two 12-bit registers and the 4-bit index counter, together with 28 tie-high cells required by the all-ones reset state. After placement, clock-tree synthesis and routing the implemented macro contains 376 instances, comprising 247 standard cells, 102 buffers inserted for timing repair, 8 clock-tree buffers and 129 fill cells. The die measures $82.59\text{ }\mu\text{m}$ by $101.31\text{ }\mu\text{m}$, an area of $8367\text{ }\mu\text{m}^2$, with a core area of $4834\text{ }\mu\text{m}^2$ at a standard-cell utilisation of 83 %.

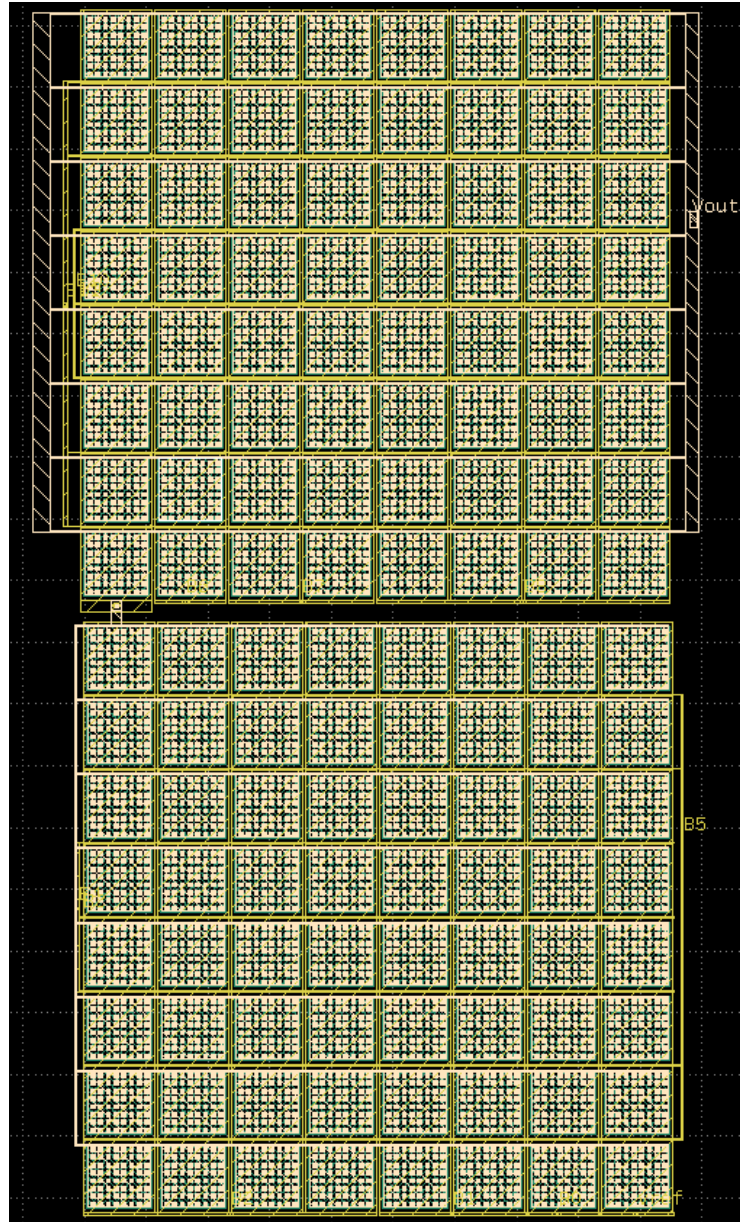


Figure 5.9 – Layout of one capacitive DAC sub-array, arranged as an 8×8 matrix of unit capacitors.

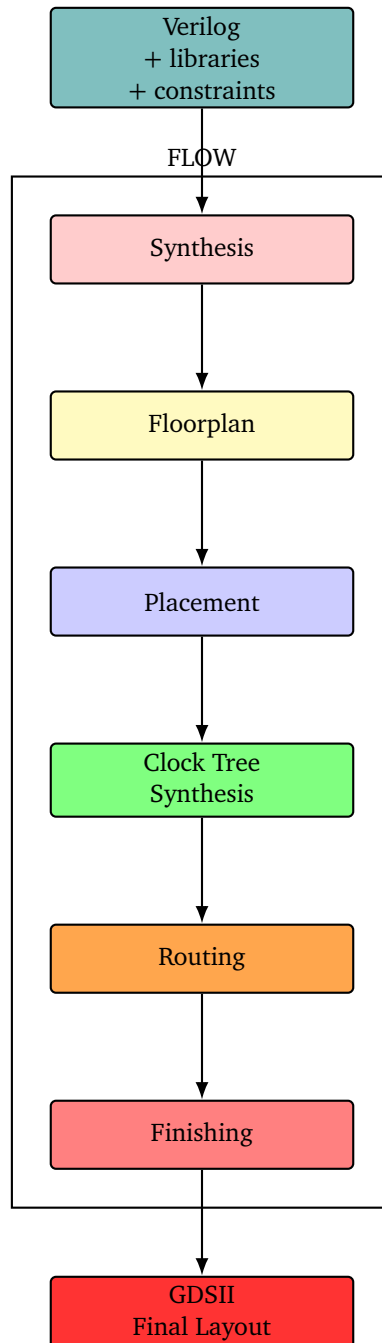


Figure 5.10 – The automated RTL-to-GDSII flow used to implement the SAR controller. Redrawn after the OpenROAD flow description [29].

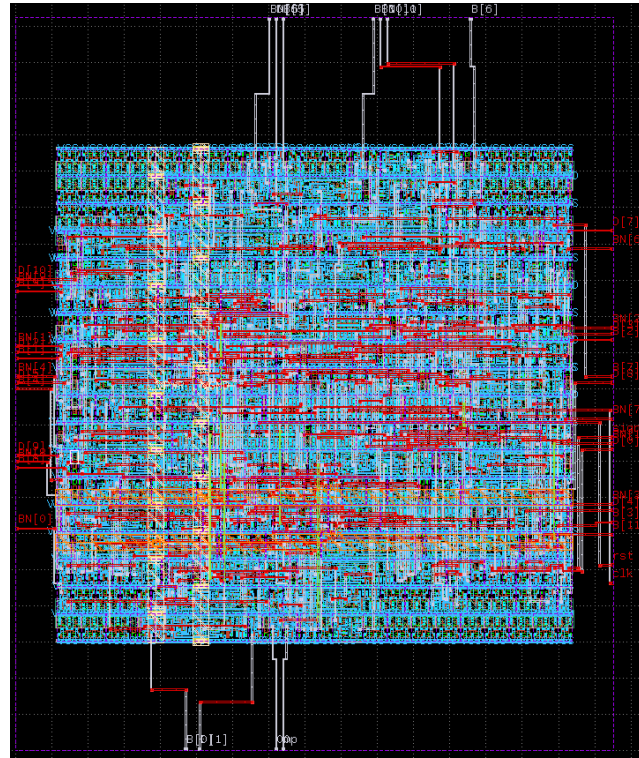


Figure 5.11 – Layout of the hardened SAR controller macro.

Timing was signed off at three corners spanning the supply and temperature range, with the results of Table 5.3. All three close with substantial margin: the worst setup slack, at the slow corner, is 6.02 ns against the 10 ns period, and the worst hold slack, at the fast corner, is 0.21 ns. There are no setup or hold violations, no maximum-slew, fanout or capacitance violations, and the clock skew is approximately 0.25 ns at every corner.

Routing completed with a total wirelength of 5326 μm and 1308 vias, all single-cut, after three optimisation iterations that resolved the 38 routing violations present in the first pass. The final routed macro has no design-rule violations and no antenna violations, so no antenna diodes were required. Analysis of the power grid gives a worst-case IR drop of 0.63 mV, negligible against the 1.2 V supply.

The estimated power of the controller at the typical corner is 187 μW , of which 150 μW is internal switching of the cells themselves, 36 μW is net switching and leakage contributes under 0.1 μW .

Table 5.3 – Timing signoff of the SAR controller macro across PVT corners.
Slack figures are worst-case.

Corner	Setup (ns)	Hold (ns)	Skew (ns)	Setup viol.	Hold viol.
Fast, 1.32 V, -40°C	7.01	0.208	0.251	0	0
Typical, 1.20 V, 25°C	6.66	0.437	0.251	0	0
Slow, 1.08 V, 125°C	6.02	0.842	0.251	0	0

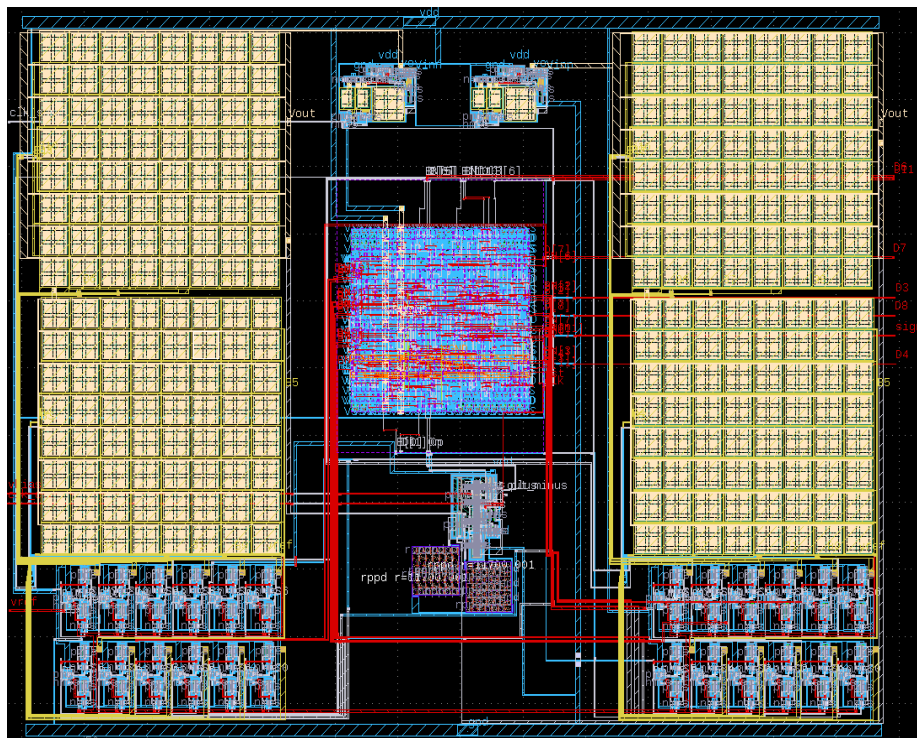


Figure 5.12 – Layout of the complete converter, with the two capacitor arrays, the analog blocks and the hardened digital controller.

5.3.5 Top-Level Integration

The complete converter is shown in Figure 5.12. The hardened controller macro is placed alongside the analog blocks, with the two capacitor arrays and their switch arrays forming the bulk of the area. The design is intended for use as a macro rather than as a standalone chip, so it carries no pad ring or input/output frame.

5.4 Verification

5.4.1 Functional Verification

The design-rule and layout-versus-schematic checks below confirm that the layout matches the schematic, but not that the converter produces the correct output code. Functional behaviour was therefore checked separately in simulation by applying known differential inputs and inspecting the digital output together with the internal control signals.

Figure 5.13 shows the conversion of a differential input of $V_{in} = V_{inp} - V_{inn} = 0.8\text{V} - 0.4\text{V} = 0.4\text{V}$. The comparator is strobed once per bit by `clk_cmp`; on each strobe the successive-approximation register clears the bit under test according to the comparator decision, following the sequence of Section 4.6, and after twelve strobes the twelve-bit magnitude and the sign bit are presented at the output. The recovered magnitude code is 1369 (0101 0101 1001). Against the ideal transfer relation of Section 4.2.1, a 0.4V input corresponds to a code of $0.4/1.2 \times 4096 \approx 1365$; the four-code difference is within the comparator offset and integral nonlinearity characterised in Chapter 6, confirming that the conversion sequence operates correctly.

Figure 5.14 shows the complementary case, $V_{inp} = 0.4\text{V}$ and $V_{inn} = 0.8\text{V}$, a differential input of -0.4V . The first comparison now resolves the opposite polarity, so the sign bit is set while the magnitude code is recovered on the negative-side register, confirming the sign-magnitude coding of Section 4.6.

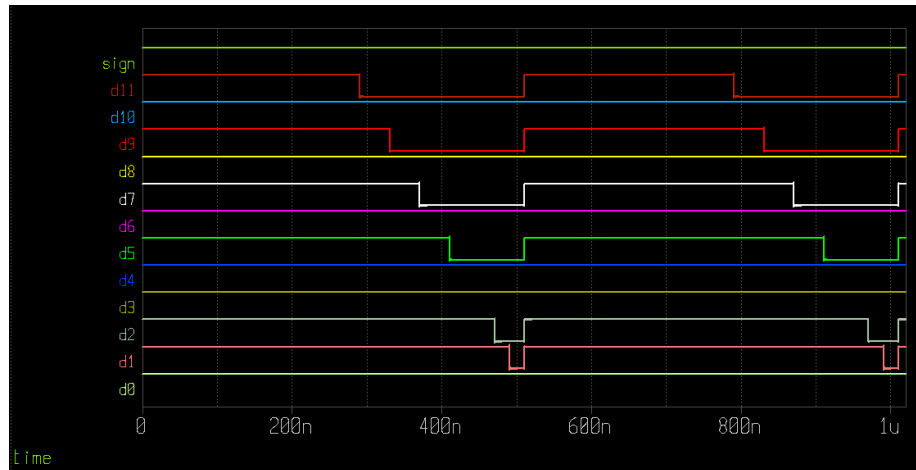


Figure 5.13 – Digital output waveform for $V_{inp} = 0.8\text{V}$, $V_{inn} = 0.4\text{V}$ ($V_{in} = 0.4\text{V}$). The comparator strobe, the successive-approximation register and the recovered twelve-bit magnitude code (1369) and sign bit are shown for a single conversion.

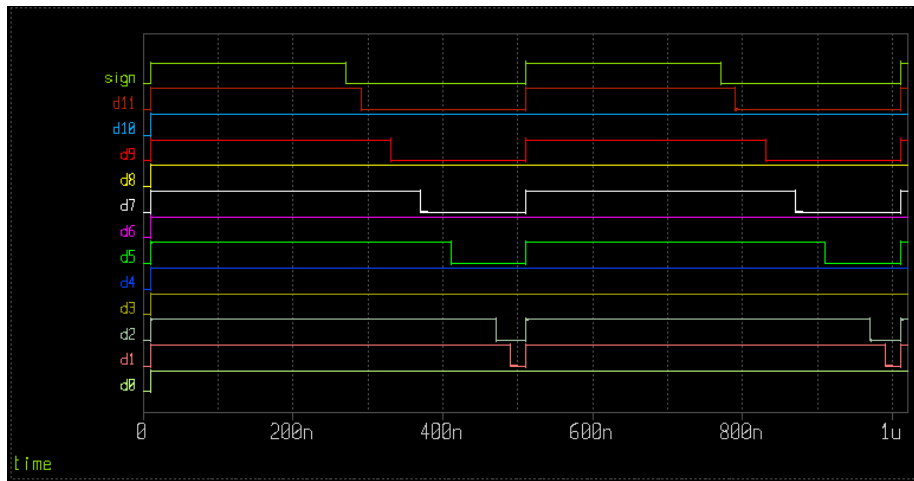


Figure 5.14 – Digital output waveform for $V_{inp} = 0.4V$, $V_{inn} = 0.8V$ ($V_{in} = -0.4V$). The first comparison resolves the opposite polarity, setting the sign bit.

5.4.2 Physical Verification

Netlist	Schematic	Cross Reference	Log
Circuits	Objects	Layout	Reference
▶ SAR_AD	▶ SAR_ADC	SAR_ADC	SAR_ADC
	▶ bootstrap_switc	bootstrap_switch	BOOTSTRAP_SWITCH
	▶ bootstrap_switc	bootstrap_switch\$1	BOOTSTRAP_SWITCH\$1
	▶ cdac ↔ CDAC	cdac	CDAC
	▶ cdac\$2 ↔ CDAC	cdac\$2	CDAC\$2
	▶ dynamic_compa	dynamic_comparator_DYNAMIC_COMPARATOR_NEW	
	▶ sar_logic ↔ SAR	sar_logic	SAR_LOGIC
	▶ switch_array_ne	switch_array_new	SWITCH_ARRAY_NEW
	▶ switch_array_ne	switch_array_new\$2	SWITCH_ARRAY_NEW\$2

Figure 5.15 – Layout-versus-schematic cross-reference for the complete converter. Every block is matched between layout and reference netlist, and the top-level cell matches.

Each block was verified individually during development and the complete converter was then checked at top level with the DRC and LVS decks supplied with the PDK. Because the top-level check operates on the full flattened layout, it covers every shape inside each block as well as the interactions between them, and so subsumes the individual block checks.

The layout-versus-schematic cross-reference of Figure 5.15 shows both bootstrapped switches, both capacitor arrays, both switch arrays, the comparator and

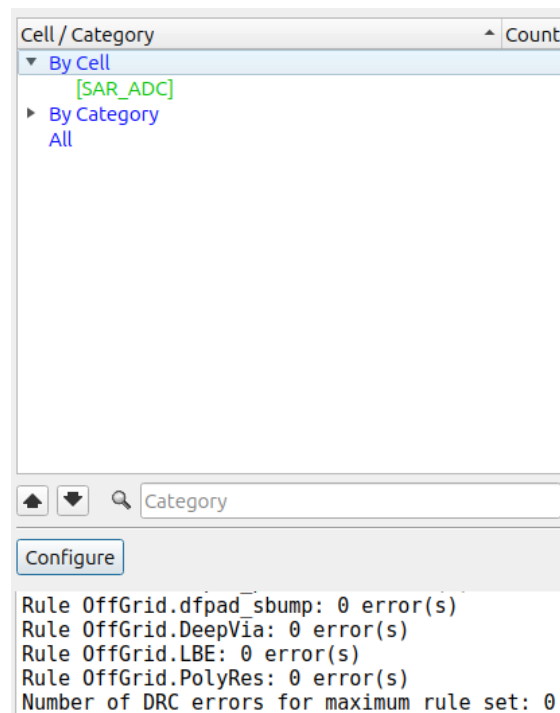


Figure 5.16 – Design-rule check of the complete converter, reporting zero violations across all rule categories.

the hardened controller each matched against their reference netlists, with the top-level cell matching and no device, net, pin or property differences. The design-rule check of Figure 5.16 reports zero violations across all rule categories. The hardened controller macro was independently signed off within the LibreLane flow, which likewise reports no design-rule violations from either the Magic or KLayout checkers, no illegal overlaps, and no layout-versus-schematic differences.

5.5 Known Limitations

Three limitations of the implementation should be stated plainly. The first two were identified late in the design cycle, after the comparator had been verified to meet its requirements across the conversion; since the converter met its performance targets, neither was judged to warrant a redesign within the scope of this work.

The tail device of the preamplifier does not operate in saturation at the nominal bias. Simulation gives $|V_{DS}| = 114 \text{ mV}$ against $|V_{Dsat}| = 230 \text{ mV}$, placing the device in the linear region, where its output impedance is lower than intended. The bias current therefore varies somewhat as the input common-mode falls through the conversion, and with it the preamplifier gain. The converter meets its performance

targets nonetheless, but a tail device biased properly into saturation would give a more constant gain across the conversion.

The preamplifier gain of 2.2 is modest and the load resistor was chosen conservatively. The input devices operate with roughly 0.78 V of saturation margin, so the load could have been raised substantially before the input pair approached the edge of saturation, giving a proportionally higher gain and a correspondingly larger reduction of the latch's input-referred offset.

The design is implemented as a macro rather than a complete chip. It has no pad ring, no electrostatic-discharge protection and no input/output frame, and metal fill has not been applied. These steps would be required before submission to a fabrication run and are outside the scope of this work.

Chapter 6

Evaluation

This chapter reports the measured performance of the converter: its static linearity, its dynamic performance, the comparator offset, and its power and figures of merit.

The static linearity was characterised from a slow input ramp applied in 1 mV steps, giving 1160 sample points across the input range, from which the transfer curve, INL and DNL were computed against a best-fit line, following the standard definitions for analog-to-digital converter testing [9]. The dynamic performance was evaluated from a 128-point discrete Fourier transform of the output for a coherent sinusoidal input of 171.875 kHz at 2 MS/s. The comparator offset was characterised by a Monte Carlo analysis of over 300 runs using the device mismatch models supplied with the PDK.

The dynamic and static performance figures reported in this chapter were obtained from schematic-level (pre-layout) simulation. The laid-out design was verified separately: it passes layout-versus-schematic checking, confirming that the extracted layout matches the schematic device for device, and post-layout functional simulation of the extracted netlist reproduces the same output codes as the pre-layout design. Because the extracted netlist carries no interconnect parasitics, it is electrically equivalent to the schematic netlist, so the two agree as expected.

Full parasitic (RC) extraction was not carried out: the open-source extraction path does not robustly handle a design of this size and complexity, and a verified parasitic-annotated netlist could not be produced within the project timeline. The figures reported here therefore reflect the drawn devices and their connectivity but not the resistance and capacitance of the interconnect. They should be read as an upper bound on performance, since interconnect parasitics would degrade the dynamic figures to some degree.

6.1 Static Linearity

The static transfer characteristic was obtained from a slow input ramp covering the full-scale range, giving 1160 output codes spanning code 64 to code 4029. A best-fit line through the transfer curve gives a gain of 3420 codes/V, against an ideal of 3413 codes/V, and an offset of -4.8 codes at zero input. The differential and integral nonlinearity, referred to this best-fit line, are shown in Figure 6.1.

The differential nonlinearity lies within -0.41 and $+0.17$ LSB, a peak magnitude of 0.41 LSB, and no missing codes or code reversals occur, so the converter is monotonic. The integral nonlinearity lies within -1.16 and $+1.13$ LSB, a peak magnitude of 1.16 LSB. The DNL, comfortably below one LSB, confirms that the unit-capacitor sizing of Section 4.2.1 keeps the local step errors small; the slightly larger INL reflects the accumulated weighting error across the array, including the deterministic contribution of the plain-unit bridge capacitor discussed in Section 4.2.2.

The shape of the INL curve is informative. The mean of the INL shifts upward at mid-scale, near $V_{in} = 0.6$ V, which is the boundary between the two 6-bit sub-arrays, the point at which the most significant bit is switched. This is precisely where the weighting error of the plain-unit bridge capacitor of Section 4.2.2 is expected to appear: the bridge attenuates the LSB sub-array by a slightly incorrect factor, so the two halves of the transfer characteristic carry a small relative offset with respect to each other, visible here as a step in the INL baseline at the sub-array boundary. The effect is deterministic and bounded, as anticipated when the trade was made, and it keeps the peak INL within 1.16 LSB. The differential nonlinearity shows no such systematic feature: it consists of dense small deviations within roughly ± 0.2 LSB, with isolated codes reaching -0.41 LSB, consistent with random capacitor mismatch rather than a structural error, and no code reversals occur.

6.2 Dynamic Performance

The dynamic performance was evaluated by applying a near-full-scale sinusoidal input at 171.875 kHz, sampling at 2 MS/s, and taking the Fourier transform of the output. The resulting spectrum is shown in Figure 6.2. The converter achieves a signal-to-noise-and-distortion ratio (SNDR) of 69.5 dB and a spurious-free dynamic range (SFDR) of 78.3 dB relative to full scale, with total harmonic distortion of -80.4 dB. The effective number of bits follows from the SNDR as [9]

$$\text{ENOB} = \frac{\text{SNDR} - 1.76}{6.02} = \frac{69.5 - 1.76}{6.02} \approx 11.3 \text{ bits.} \quad (6.1)$$

The distortion is dominated by the third harmonic; the second harmonic is far lower, at approximately -100 dB. This pattern is characteristic of a well-balanced differential converter. The differential structure cancels the even-order distortion

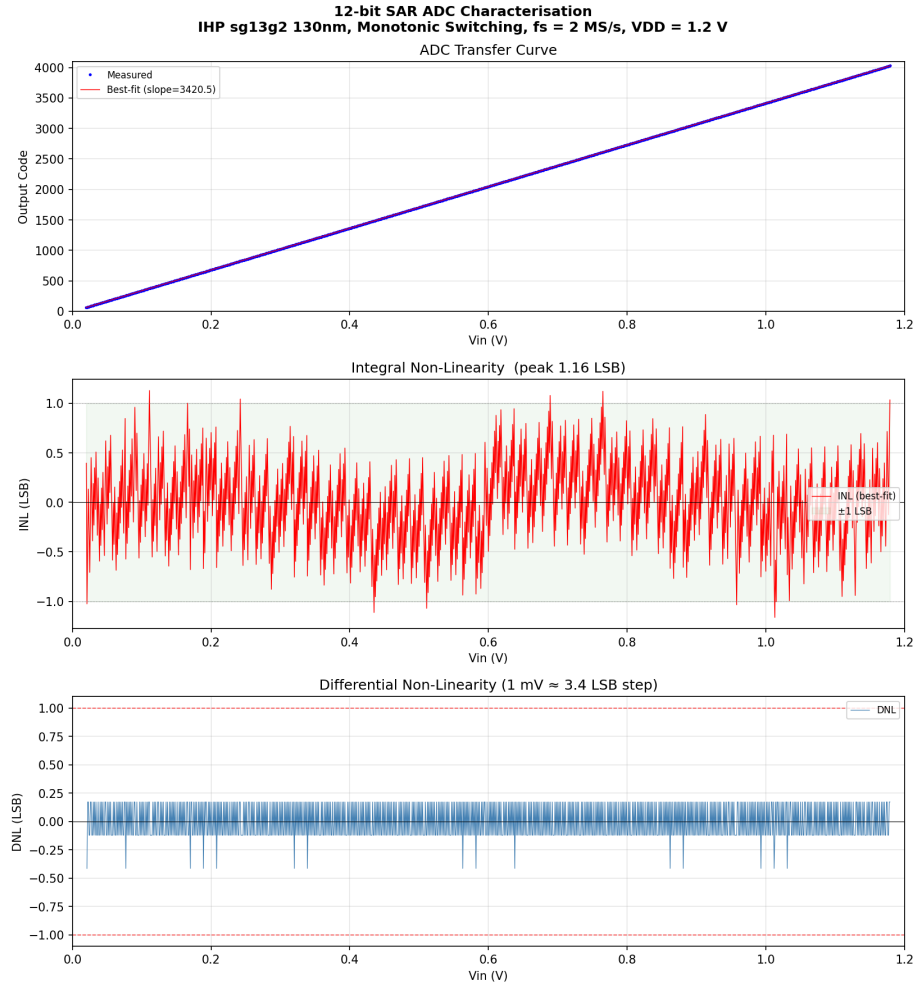


Figure 6.1 – Differential and integral nonlinearity of the converter, extracted from a ramp of 1160 samples over the input range.

terms, which is why the second harmonic is suppressed and why the layout symmetry of the two halves matters [8]; what remains is the odd-order distortion, appearing as the third harmonic. Its most likely source is the residual signal dependence of the sampling-switch on-resistance, which is symmetric about mid-scale and therefore produces odd-order rather than even-order terms. The bootstrapped switch of Section 4.5 holds the on-resistance nearly constant but not perfectly so, and the residual variation sets the distortion floor. The same limitation is reported by Zhu et al., who attribute their high-frequency distortion to the sampling switches [19].

The record length for the dynamic analysis is 128 points. This is short for a 12-bit converter: the processing gain of a 128-point transform is limited, which raises the noise floor of the spectrum and bounds the precision to which the SNDR and SFDR

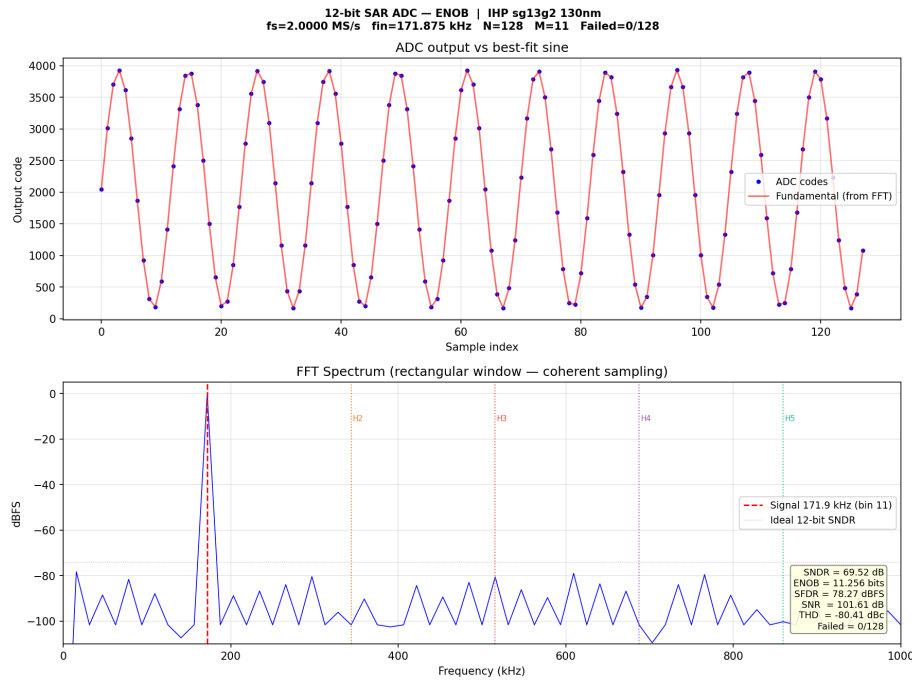


Figure 6.2 – Output spectrum for a 171.875 kHz full-scale sinusoidal input sampled at 2 MS/s.

can be resolved. The dynamic figures should therefore be read as indicative; a longer coherent record would refine them, and this was not run within the available time.

6.2.1 Ideal vs. Achieved

It is worth comparing the achieved dynamic range against the theoretical limit. For an ideal 12-bit converter the signal-to-noise ratio is $6.02 \times 12 + 1.76 = 74.0$ dB, corresponding to the full 12 bits. The measured SNDR of 69.5 dB is 4.5 dB below this, a loss of about 0.7 bit. This shortfall is accounted for by the non-idealities characterised elsewhere in this chapter: the comparator contributes noise and a finite offset, the residual capacitor weighting error appears as the integral nonlinearity above, and the third-harmonic distortion from the sampling switch reduces the signal-to-noise-and-distortion ratio directly. The converter therefore delivers 11.3 effective bits of its 12 nominal bits.

6.3 Comparator Offset

Because a comparator offset shifts every code transition by the same amount, it appears as an offset of the whole converter within a given instance and does not

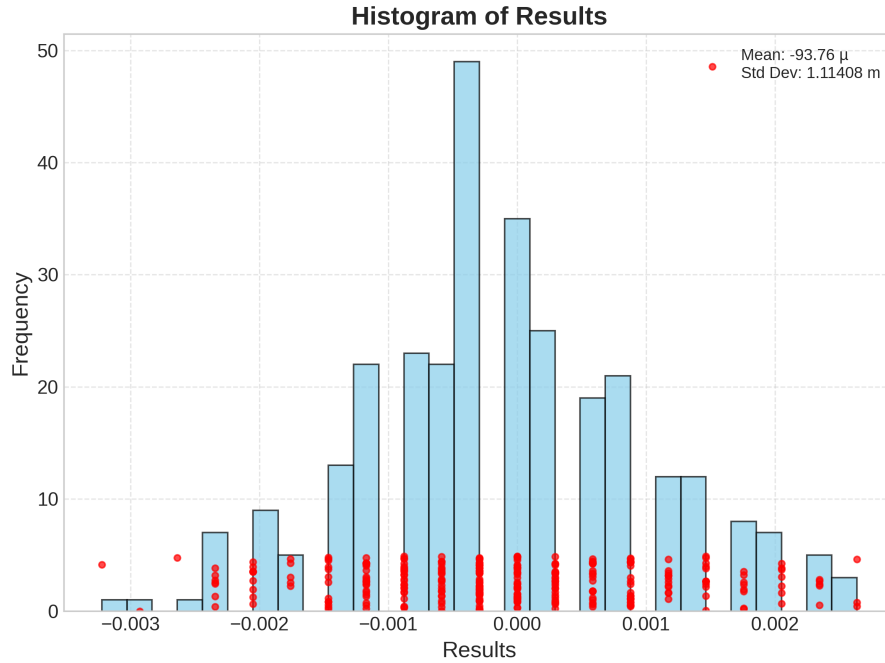


Figure 6.3 – Monte Carlo distribution of the comparator input-referred offset over 300 runs. The standard deviation is approximately 1.11 mV, corresponding to 3.9 LSB at the 293 μV LSB of Equation (4.1)

by itself degrade linearity. Its magnitude varies from instance to instance with device mismatch, however, and was characterised by Monte Carlo simulation. The input-referred offset has a standard deviation of approximately 3.9 LSB, shown in Figure 6.3. This is consistent with the modest preamplifier gain reported in Section 5.2.3; a higher gain would divide the latch offset by a larger factor and reduce it, as noted in the limitations of that chapter.

6.4 Power and Figures of Merit

The converter draws 185.5 μA from the 1.2 V supply and 41.7 μA from the reference, with negligible bias and clock currents, for a total power of

$$P = (185.5 \mu\text{A} + 41.7 \mu\text{A}) \times 1.2 \text{V} \approx 272 \mu\text{W}. \quad (6.2)$$

The reference current is included, since it is real power drawn from the supply to drive the array. The reference current accounts for roughly 18 % of the total power, a relatively small fraction, consistent with the low switching energy of the monotonic scheme chosen in Section 4.3.

The two standard figures of merit follow from this power together with the dynamic performance above, both taken from the same simulation. The Walden

figure of merit is [10]

$$\text{FOM}_W = \frac{P}{2^{\text{ENOB}} \cdot f_s} = \frac{272 \mu\text{W}}{2^{11.3} \times 2 \text{ MS/s}} \approx 54 \text{ fJ/conv-step}, \quad (6.3)$$

and the Schreier figure of merit is [12]

$$\text{FOM}_S = \text{SNDR} + 10 \log_{10} \left(\frac{f_s/2}{P} \right) = 69.5 + 10 \log_{10} \left(\frac{1 \text{ MHz}}{272 \mu\text{W}} \right) \approx 165 \text{ dB}. \quad (6.4)$$

6.5 Summary and Comparison

Table 6.1 summarises the measured performance. These figures are placed against comparable converters in Table 3.1 of Chapter 3, which shows that the converter is competitive in resolution and efficiency with recent open-source and low-power SAR designs while being realised entirely on an open toolchain.

Table 6.1 – Summary of measured performance.

Parameter	Value
Technology	IHP SG13G2, 130 nm
Supply	1.2 V
Resolution	12 bit
Sample rate	2 MS/s
ENOB	11.3 bit
SNDR	69.5 dB
SFDR	78.3 dB
THD	−80.4 dB
DNL (peak)	0.41 LSB
INL (peak)	1.16 LSB
Power	272 μW
FOM_W	54 fJ/conv-step
FOM_S	165 dB

Chapter 7

Conclusion

7.1 Summary

This thesis has presented the design of a 12-bit successive-approximation-register analog-to-digital converter in the IHP SG13G2 130 nm open PDK, carried out entirely with a free and open-source toolchain. The converter is fully differential and uses monotonic switching, with a split 6+6 binary-weighted MIM capacitor array, a bootstrapped sampling switch, a resistively-loaded preamplifier followed by a StrongARM latch as the comparator, and a synchronous SAR controller implemented through an automated digital flow.

The work followed the design through from architecture to layout. Each block was chosen and justified against the requirements of a 12-bit converter: the split array to keep the capacitance and area manageable, the monotonic scheme for its low switching energy and simple control, the bootstrapped switch for linear sampling, the PMOS-input comparator to accommodate the falling common-mode that the monotonic scheme produces, and a sign-magnitude output coding that follows naturally from the first comparison. The unit capacitor was sized by simulation against both the sampling-noise bound and the weighting accuracy of the array, and the analog blocks were laid out with attention to matching, most notably the common-centroid placement of the two most significant capacitor weights. The digital controller was described in SystemVerilog and hardened with LibreLane, closing timing at all process corners.

The complete converter passes design-rule and layout-versus-schematic checks, and pre-layout simulation gives an effective resolution of 11.3 bits, an SNDR of 69.5 dB, peak DNL and INL of 0.41 LSB and 1.16 LSB, and a total power of 272 μ W at 2 MS/s, corresponding to a Walden figure of merit of 54 fJ per conversion step. The measured integral nonlinearity confirms the expected behaviour of the design, with the deterministic weighting error of the bridge capacitor appearing precisely

at the sub-array boundary, and the low second-harmonic distortion confirms the balance of the differential structure.

7.2 Contributions

The principal outcome of this work is a complete, verified 12-bit SAR ADC design realised end to end on an open-source flow and open PDK, from schematic and custom analog layout through to a hardened digital controller and a design-rule- and IVS-clean top-level macro. The design demonstrates that a converter of this resolution can be taken through the full custom-analog and digital design path using only freely available tools on the SG13G2 process.

Alongside the converter itself, the work contributes a documented account of the design decisions and their trade-offs at each block, including the sizing of the unit capacitor against noise and matching, the deliberate use of a plain-unit bridge capacitor and its measured consequence, and the addition of a static preamplifier to the dynamic comparator with its associated gain and power trade-off.

7.3 Limitations and Future Work

Several directions follow directly from the limitations identified in Chapters 5 and 6.

The most immediate limitation is the lack of full parasitic extraction. The scope of this thesis could not progress beyond post-layout simulation without interconnect parasitics because a verified RC-extracted netlist could not be generated using the open-source design flow within the project timeline. Confirming the performance with a parasitic-annotated simulation, or through measurement of a fabricated device, is the natural next step, and would establish how much the dynamic figures degrade once interconnect is included.

At the circuit level, two improvements were identified. The preamplifier tail device does not operate in saturation at the nominal bias, so a revised bias or device sizing would give a more constant gain across the falling common-mode. The preamplifier gain itself is modest and was shown to leave substantial saturation margin, so a larger load resistor would raise the gain and reduce the comparator's input-referred offset. Both are contained changes that would improve the dynamic performance without altering the architecture.

Beyond the converter itself, the design is delivered as a macro rather than a complete chip. Integrating it with a pad ring, electrostatic-discharge protection and metal fill, and submitting it to a fabrication run, would allow the simulated performance to be verified against silicon and would complete the path from open PDK to manufactured device.

Glossary

ADC Analog-to-Digital Converter

BiCMOS Bipolar Complementary Metal-Oxide-Semiconductor

CDAC Capacitive Digital-to-Analog Converter

CMOS Complementary Metal-Oxide-Semiconductor

DAC Digital-to-Analog Converter

DNL Differential Nonlinearity

DRC Design-Rule Check

ENOB Effective Number of Bits

FOM Figure of Merit

GDSII Graphic Design System II (layout stream format)

INL Integral Nonlinearity

LSB Least-Significant Bit

LVS Layout-Versus-Schematic

MIM Metal-Insulator-Metal

MSB Most-Significant Bit

PDK Process Design Kit

PVT Process, Voltage, Temperature

RTL Register-Transfer Level

SAR Successive-Approximation Register

SFDR Spurious-Free Dynamic Range

SNDR Signal-to-Noise-and-Distortion Ratio

SNR Signal-to-Noise Ratio

THD Total Harmonic Distortion

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Appendix A

SAR Controller Source

The synchronous SAR controller described in Section 4.6 and implemented in Section 5.3.4 is given here in full, together with the LibreLane configuration used to harden it.

A.1 SystemVerilog Description

```
1 module sar_logic (
2     input logic      clk,
3     input logic      rst,
4     input logic      Op,
5     input logic      Om,
6     output logic [11:0] B,
7     output logic [11:0] BN,
8     output logic      sign,
9     output logic [11:0] D
10 );
11
12     logic [3:0] dbit;
13
14     always_ff @(posedge clk) begin
15         if (rst) begin
16             B      <= 12'd4095;
17             BN     <= 12'd4095;
18             dbit   <= 4'd11;
19         end
20         else if (Om ^ Op) begin
21             if (Om == 1'b1)
22                 BN[dbit] <= 1'b0;
23             else
24                 B[dbit] <= 1'b0;
25
26             if (dbit > 0)
```

```
27         dbit <= dbit - 1;
28     end
29 end
30
31 logic [11:0] rot_pos;
32 logic [11:0] rot_neg;
33
34 assign rot_pos = {B[10:0], B[11]};
35 assign rot_neg = {BN[10:0], BN[11]};
36
37 assign sign = BN[11];
38 assign D = (B[11]) ? rot_pos : rot_neg;
39
40 endmodule
```

Listing A.1 – SAR Logic Module

A.2 LibreLane Configuration

Listing A.2 presents the LibreLane configuration file used for synthesis and physical implementation of the SAR controller.

```
1 DESIGN_NAME: sar_logic
2
3 VERILOG_FILES:
4   - dir::sar_logic.sv
5
6 CLOCK_PORT: clk
7 CLOCK_PERIOD: 10
8
9 STD_CELL_LIBRARY: sg13g2_stdcell
```

Listing A.2 – LibreLane configuration (config.yaml)

Appendix B

Top-Level Schematic

Figure B.1 shows the complete converter as implemented, with the analog blocks described in Section 5.2 and the synthesised SAR controller of Section 5.3.4 shown together in a single view. The two switch arrays are driven by the B and BN register outputs respectively, realising the differential monotonic switching described in Section 4.3.

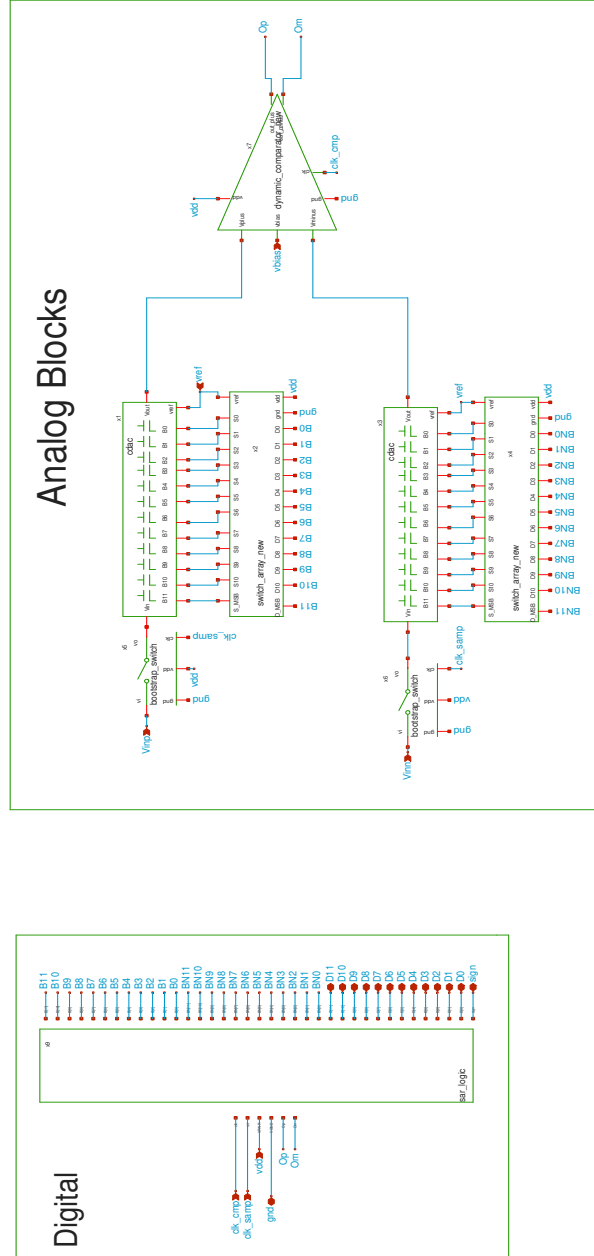


Figure B.1 – Top-level schematic of the converter. The analog blocks (right) comprise the two bootstrapped sampling switches, the two split capacitive DACs with their switch arrays, and the shared dynamic comparator. The synchronous SAR controller (left) receives the comparator outputs O_p and O_m and drives the two switch arrays through the register outputs B and BN , presenting the converted code on D together with the sign bit. The digital and analog parts are drawn separately for clarity; the signals shared between them are connected by name.

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